

Five Facts About Inflation Expectations: Evidence from Four Different Groups of Agents*

Eleonora Granziera^{†1}, Lovisa Reiche²,
Nicolò Maffei-Faccioli³, Michael Weber⁴, and Tuva Fastbø⁵

^{1,3,5}Norges Bank

²BI Norwegian Business School

⁵Purdue University Daniels School of Business

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Abstract

We provide five facts about the inflation expectations of households, firms, social partners, and economists from harmonized data spanning more than two decades: (i) Households' mean expectations were highest pre-inflation surge, but firms' expectations were highest since 2021 due to differential energy price movements. (ii) Disagreement displays similar patterns. (iii) Short-term shape medium-term forecasts, with magnitudes differing across groups and regimes. (iv) The pass-through from inflation to wage expectations is below one, strongest for business leaders and weakest for social partners. (v) Electricity price shocks causally raise inflation expectations. Taken together, the unanchoring of inflation expectations is pervasive across time and groups but the threat of price-wage spirals remains limited.

Keywords: Inflation Expectations, Wage Expectations, Households, Firms, Professional Forecasters, Unions.

JEL codes: D84, E31, E52.

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[†]Corresponding author; Email: eleonora.granziera@norges-bank.no.

1 Introduction

Subjective expectations of different groups of agents play a central role in modern macroeconomic models. For instance, households’ inflation expectations influence their perceived real returns on savings, thereby shaping their current consumption decisions (Bachmann et al. 2015, Coibion et al. 2022, D’Acunto et al. 2022). Similarly, firms’ beliefs about inflation will determine their pricing strategies, labor demand, investment, and financing choices (Coibion et al. 2018, Ropele et al. 2022) and, consequently, current inflation. The inflation expectations of employer organizations and trade unions (the social partners) influence nominal wage setting. While a growing literature studies how households and firms form inflation expectations alongside professional forecasters,¹ no systematic evidence exists on (a) the similarities and differences between the expectations of households, firms, and economists over time and across economic environments and (b) the expectations of social partners.

We address these gaps by using unique surveys of Norwegian households, business leaders, social partners and economists conducted quarterly since 2002 to provide robust facts across agents and over time that can inform theorists and applied macroeconomists when developing models that are informed by empirical facts. The Norges Bank Expectations’ Survey (NBES) includes data on inflation and wage expectations across various horizons and has two distinct and novel features. First, it is harmonized, ensuring all agents are surveyed during the same time period using identical questions, under a unified survey design. This harmonization facilitates direct comparisons by exposing all respondents to the same macroeconomic environment and the same information set.² Crucially, all agents face similar question designs and forms, so that questionnaire features do not affect survey responses (Becker et al. 2023, Bruine de Bruin et al. 2011, D’Acunto et al. 2023). This feature differs from the comparisons presented in Candia, Coibion & Gorodnichenko (2023) for which households and firms are interviewed independently of each other with often different forecast horizons, types and modes of elicitation, wordings, and localities with heterogeneous institutional settings. Second, the NBES uniquely includes social partners – a critical group given their significant role in wage setting and economic coordination – and economists from academia and the financial sector, offering an unparalleled and comprehensive view of economic expectations. Spanning over two decades, this dataset covers a significantly longer time horizon than most existing surveys, allowing for consistent and rigorous comparisons across agents and over periods characterized by potentially different macroeconomic environ-

¹Bachmann et al. 2023 provides a recent overview of this fast growing body of work.

²The surveys are in the field over a period of three weeks that does not coincide with releases of inflation, national accounts or Norges Bank’s policy rate decisions. We provide more details in the next Section.

ments, such as periods of low and high inflation.³ The survey meets all the criteria outlined in Coibion et al. (2020) for achieving high quality, benefiting from its large and representative cross-sections, high frequency, long time span, as well as the quantitative formulation of its questions.

Leveraging these unique features, we revisit five stylized facts about inflation expectations, offering new insights into how different agents form and adjust their expectations over time. In particular, we highlight that these facts are not necessarily static, but vary in response to changing macroeconomic conditions, notably in periods of low versus high inflation. The five facts are:

(1) **Level Differences.** The literature suggests firms’ expectations typically fall between those of professional forecasters and households (Candia, Coibion & Gorodnichenko 2023), with households’ and firms’ average expectations systematically exceeding median expectations due to a long right tail (D’Acunto et al. 2023). Our results confirm the findings of Candia, Coibion & Gorodnichenko (2023) for the period spanning 2002–2021 but reveal a notable shift thereafter. Following the inflationary period in 2021, firms’ inflation expectations, both short- and medium-term, exceeded those of households and remained higher until Q4 2024, the last survey wave in our sample. Firms and households faced different energy-price shocks due to partial energy subsidies for households, suggesting differences in inflation expectations across agents may depend on both the overall inflation level and the specific shocks they are exposed to, consistent with the evidence for households (D’Acunto, Malmendier, Ospina & Weber 2021), though evidence for firms remains limited. Furthermore, social partners form expectations similarly to economists, but their expectations became more skewed in the post-2021 period. During times of heightened inflation, trade unions (representing employees) raise their inflation expectations by more than employer organizations, though in most periods, both groups exhibit similar expectations.

(2) **Cross-Sectional Disagreement.** Under full-information rational expectations (FIRE), all agents form identical inflation expectations and no dispersion should arise; however, empirical evidence shows substantial disagreement (Mankiw et al. 2003). Cross-sectional dispersion of expectations is typically largest for households, followed by firms and professional forecasters (Candia, Coibion & Gorodnichenko 2023). It is time-varying for all agents (Fofana et al. 2024), and systematically related to household and firm characteristics (Baumann et al. 2024, Bruine De Bruin et al. 2010, Candia, Weber, Gorodnichenko & Coibion 2023, D’Acunto et al. 2023, Piccolo et al. 2025, Reiche 2025). So far, no systematic evi-

³The inflation environment can shape the attention agents pay to inflation, see Bracha & Tang (2025), Cavallo et al. (2017), Korenok et al. (2023), Pfäuti (2023), Weber et al. (2025).

dence exists on whether the dispersion in forecasts differs across different subsets of experts (Clements et al. 2023) and no evidence exists regarding the disagreement among social partners. Until 2021, the disagreement among households, firms and professional forecasters in Norway resembles that of other surveys. Similarly to the level of inflation expectations, the disagreement among firms rises substantially after 2021 and even exceeds the disagreement among households. Similar to previous work, gender, income and age systematically impact households' inflation expectations. For firms, we find large heterogeneity across sectors driven mostly by expectations of sales and purchasing prices. Both professional forecasters and social partners show low levels of dispersion throughout the sample.

(3) **Term Structure.** A growing body of evidence suggests inflation expectations of both households and firms are largely explained by their perceptions, that is, the assessment of the current inflation rate (Candia, Weber, Gorodnichenko & Coibion 2023, D'Acunto & Weber 2024, Huber et al. 2023, Weber et al. 2022). However, the relationship between short-term and longer-term inflation expectations – referred to as the term structure of inflation expectations – is less well studied. Candia, Coibion & Gorodnichenko (2023) and D'Acunto et al. (2023), D'Acunto et al. (2024) suggest both firms' and households' long-term expectations are related to short-term expectations, but evidence for other agents remains limited. Our survey includes inflation expectations at the individual level across the different groups of agents and for three time horizons: 12 months ahead (short-term), 2-3 years ahead (medium-term), and 5 years ahead (long-term).⁴ This setting allows us to analyze short-to-medium-term pass-throughs across different agents and inflation environments. For all agents, short-term expectations significantly shape medium-term expectations. During the period of low and stable inflation, the pass-through is strongest for social partners (above 1), followed by business leaders, households and economists. After 2021, when inflation surged, the pass-through increased for firms but declined for households and economists, suggesting that firms perceived inflation as more persistent, whereas other agents expected faster mean reversion. This heterogeneity in perceived persistence may play a role in how quickly inflation shocks feed into prices and wages, with potential implications for the persistence of inflation and the transmission of monetary policy. Together, these results reveal distinct patterns of expectations anchoring across agents and inflation environments.

(4) **Correlation with Wage Expectations.** Agents often form their expectations about different macroeconomic variables jointly (Andre et al. 2022, Buchheim et al. 2024, Coibion et al. 2023). For instance, firms, households and trade unions may link their wage expectations to expectations about changes in the nominal price level. This connection in-

⁴For same agents, we also observe inflation perceptions.

fluences how households perceive their real wage growth, how firms expect their costs to rise, and, ultimately, how they set prices. Some evidence points towards a weak link between firms' inflation expectations and their expected wage bills (Abberger et al. 2024, Baumann et al. 2024, Buchheim et al. 2024, Coibion et al. 2018, Savignac et al. 2024), potentially due to industry-specific expectations (Andrade et al. 2022). Similarly, the effect of inflation expectations on own income expectations for households appears low, with some differences between high- and low-income earners and men and women (Hajdini et al. 2022, Jain et al. 2024, Reiche & Maffei-Faccioli 2025). The existing evidence comes largely from one-off experiments or small samples, with Jain et al. (2024) being one exception. We explore time-variation in the pass-through of inflation expectations and document the pass-through for social partners, key actors in the wage-setting process in Norway and across European countries. Specifically, we examine how inflation expectations influence personal income expectations (for households), wage bill expectations (for firms), and wage growth expectations (for professional forecasters and trade unions). The pass-through is highest for firms and lowest for households. Whereas the pass-through is positive and significant for all agents, it remains substantially below one, limiting concerns of wage-price spirals. We also document time variation in the pass-through: during the recent inflationary surge, it increased for social partners, firms and economists, but decreased for households.

(5) **Response to energy shock.** While previous studies establish a link between household inflation expectations and oil prices (Binder 2018, Coibion & Gorodnichenko 2015, Kilian & Zhou 2022), little is known about the effect of electricity prices. In a recent paper, Patzelt & Reis (2024) estimate a 1% increase in electricity prices increases inflation expectations of European households by 1.0 to 1.3 basis points. On the firm side, Hensel et al. (2024) show inflation expectations of French firms rise in response to carbon policy shocks, whereas Wehrhofer (2023) finds a positive pass-through of electricity prices to own sales prices, but not to aggregate inflation expectations for German firms. We exploit an electricity price shock that affected Norway differently across its North and South regions. In 2021Q2, energy prices in the South began to diverge significantly from those in the North due to factors exogenous to the Norwegian economy. Prior to this event, inflation expectations of firms in the North and South were aligned but began to diverge following the shock. We estimate that, on average, over the quarters when electricity prices in the South diverged from those in the North, firms in the South had 12-month-ahead inflation expectations that exceeded those of firms in the North by more than 1 percentage point. The response is even stronger for two-year (medium-run) expectations, suggesting that firms viewed the shock as persistent. This reaction is consistent with the broader pattern observed for the term structure, where firms' expectations exhibited higher perceived persistence than those of the other groups

during the high-inflation period. In contrast, when we repeat the analysis for households, we find no significant difference in their inflation expectations across regions. The Norwegian government’s electricity support package for households, introduced in the fourth quarter of 2021, can rationalize this pattern, because it helped keep electricity bills in the South very close to those in the North.

Documenting these stylized facts is important for several reasons. First, previous comparisons in the literature were not able to compare responses across groups of agents using the same degree of survey harmonization and did not include social partners, who play a central role in wage setting and thus in determining the inflation-wage pass-through. Note that social partners are relevant outside of the Norwegian context, as the majority of workers in OECD economies have their wages determined, directly or indirectly, through some form of collective negotiation between organized groups. Moreover, prior comparisons often focused primarily on data collected post-2020, a period marked by a very different inflationary environment. We show the time dimension of our dataset is a key factor, because stylized facts are not necessarily static, rather they evolve based on the macroeconomic environment. Indeed, many of these facts depend on the level of inflation ([Bracha & Tang 2025](#), [Korenok et al. 2023](#), [Pfäuti 2023](#), [Weber et al. 2025](#)).

The paper is organized as follows. Section 2 describes the surveys. Section 3 revisits five key stylized facts in the literature on inflation expectations and Section 4 concludes.

2 Data

The Norges Bank expectations survey is a nationally representative survey of four groups: households, business leaders, social partners and economists. The survey has been running quarterly since 2002Q1, with responses at the individual level available from 2009Q1. An external survey company administers the survey on behalf of Norges Bank.⁵

The survey is designed to be representative of the Norwegian population, and the dataset contains a limited number of characteristics about the interviewed households and firms. About 1,000 households are surveyed in each wave and interviews are conducted via telephone.⁶ Respondents are recruited randomly from Norway’s population of age fifteen and above. They are asked about their gender, age, employment status and location. With

⁵The survey has been conducted by Ipsos since 2020Q3. The survey was previously carried out by Epinion, Opinion and TNS Gallup.

⁶This sample size corresponds to interviewing one in 5,500 households compared to one in 560,000 households for the Michigan Survey of Consumers and one in 258,000 for the New York FED Survey of Consumer Expectations.

regards to the employment status, respondents can choose among student, employed in the private sector, employed in the public sector, unemployed or retired. Finally, they are asked about their region of residence: North Norway, Mid-Norway, East Norway, West Norway, South Norway. Oslo, Norway's capital and its largest city, is identified as a separate region, though it is located in Eastern Norway. About 85-90% of respondents are replaced every wave.

Firms' expectations are collected through an online survey of at least 500 respondents, consisting of CEOs, CFOs and purchasing managers in Norwegian businesses with 20 or more employees. The sample comes from a pre-recruited panel. These business leaders belong to the following sectors: manufacturing, building and construction, services and retail.⁷ Business leaders are also asked about their geographical location, with the same options as for households. Finally, firms are classified as small or large based on the number of employees, with a cut-off value of 50. Respondents stay in the survey on average for two consecutive quarters.

Economists consist of 40 experts in the financial industry and academia. The former group is therefore comparable to the forecasters included in the Survey of Professional Forecasters of the Federal Reserve Bank of Philadelphia or the European Central Bank. Economists in academia are professors of economics or finance employed by Norwegian universities.

Social partners refer to representatives of employers and employees who work in employer organizations and trade unions. The survey includes about 40 respondents and distinguishes between representatives from employer organizations and those from employee organizations. The surveyed representatives hold key positions in their organizations, such as chief economists, general secretaries and managers. For both economists and social partners, the survey mode is online and respondents stay in the survey for several waves.

All interviews are conducted over a period of about 20 days, from the middle of the first month in the quarter to the beginning of the second month. The start and end dates of the interview period are chosen such that no new CPI or National Account release or monetary policy decision takes place during the field phase. For example, in the 2024Q2 wave, respondents were surveyed between April 11 and May 2, the monetary policy decision was announced on May 3, and the National Accounts figures for 2024Q1 were released on May 16. It is important to stress that all groups of agents are interviewed over the same period, and because no major scheduled announcements about the Norwegian economy are

⁷Public, oil or financial sector are excluded from the survey.

made during the interview period, agents have access to comparable information about the domestic macroeconomic environment. Individuals are not linked from wave to wave so the dataset does not have a panel component, but rather a repeated cross-section structure.

All agents are asked about expected changes in prices twelve months ahead. The question is formulated as follows: “What do you think the general rise in prices for goods and services will be in 12 months, as measured by the 12-month change in the consumer price index (CPI)?”⁸ Thus, respondents are asked specifically about the change in the overall price level, as measured by the CPI. Other surveys ask either about the *inflation rate*, e.g., as in the New York Fed and the Bank of Canada Survey of Consumer Expectations, or about the *general increase in prices*, like in the European Central Bank Consumer Expectations Survey or the Michigan Survey of Consumers. Both alternative formulations have possible downsides: consumers might not understand the concept of inflation. When asked about prices, consumers might extrapolate from the change in prices of more recently or more frequently purchased goods (D’Acunto, Malmendier, Ospina & Weber 2021, Bruine de Bruin et al. 2012, 2011). The Norges Bank formulation refers explicitly to prices of goods and services included in the CPI basket, so even if agents are aware that different price indices exists, they are directly guided to think about overall consumer price inflation (Weber et al. 2022). All agents are asked the same questions, again facilitating comparisons across groups. The answers are not probed. The only difference across groups is that households are asked a qualitative question, that is, whether they expect prices to increase, decrease or remain unchanged, before the quantitative question, similarly to the Michigan Consumer Survey. Finally, the same question regarding expected price changes is also asked for horizons of two and five years. The latter question was introduced only in 2023Q1 for households and business leaders. Before then, respondents were asked about their inflation perceptions, namely the percentage change in prices over the past twelve months. The question on perceptions was reintroduced for business leaders in 2024Q4.

Expectations are also elicited for other variables, most notably wages. However, the definition of wages and the forecast horizons differ across groups. Households and business leaders are asked about expected growth in their personal wages or their company’s wages, respectively, whereas economists and social partners are asked about expected average wage growth in the overall economy. Whereas households report their expectations for wage growth over the past and the next twelve months, all other groups provide fixed-event forecasts, referring to the current and next year. Finally, economists and social partners are also asked about expected wage growth five years ahead.

⁸The questionnaire is available in section A of the Appendix.

Inflation in Norway follows similar dynamics as in other advanced economies. Over the sample covered by the survey, CPI All Items inflation in Norway averaged 2.4%, with a minimum of -1.3% in 2004Q1 and a maximum of 6.7% in 2022Q3. These values are comparable to those for US CPI inflation, where the average was 2.5% and the minimum and maximum were -1.6% and 8.6%, respectively.

3 Five stylized facts for four groups of agents

In this section, we document five facts about the inflation expectations of different groups of agents using the NBES surveys.

3.1 Differences in Level

We start by documenting differences in the level of inflation expectations across different types of agents and time horizons. Figure 1a shows the average 12-month-ahead (short-term) inflation forecast for households, business leaders, economists and social partners made in quarter t , together with the year-over-year CPI inflation realized in that same quarter, i.e. the inflation rate between quarter $t - 4$ and t . The expectations of all agents co-move with actual CPI inflation (grey line). On average, economists have the lowest average expectations, followed closely by social partners.⁹ Among economists, those based in academic institutions tend to report higher inflation forecasts, especially since 2021, see Figure 1b. Similarly, when separating social partners in employer organizations and trade unions, the latter, which represent employees, tend to report higher forecasts, especially during periods of elevated inflation. When inflation is low, no systematic difference between the two groups arises, as highlighted in Figure 1b.

For households and business leaders, the ordering is less clear. Until 2021Q1, households' expectations exceeded those of business leaders. The gap narrowed after 2015, possibly due to a slight change in the survey design. From 2015 onward, the question for households directly asked about price changes for CPI inflation, whereas previously it referred to general prices (business leaders, however, were always asked about CPI inflation). This change in wording has the potential to lower households' inflation expectations on average (Bruine de Bruin et al. 2011, 2012, 2017). Since 2021Q1, business leaders' expectations have increased beyond those of households.

After discussing differences in the levels of short-term inflation forecasts, a natural

⁹The two-year-ahead (medium-term) inflation forecasts in Figure C.1 in the Appendix display similar patterns. For households, the question specifies a forecast horizon of 2-3 years ahead.

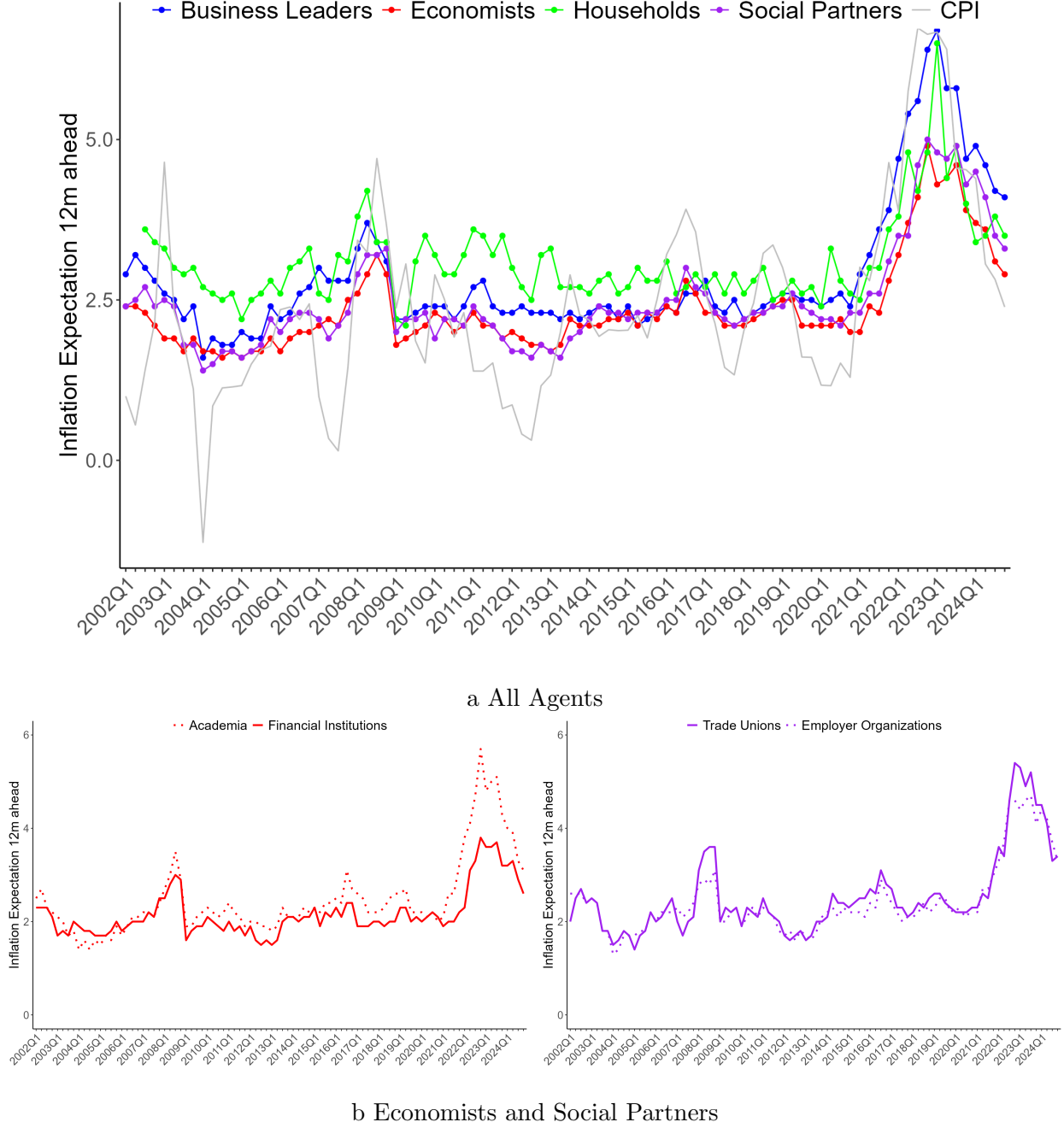


Figure 1: Timeseries of average inflation expectations (12 months ahead) since 2002
Note: The bottom left hand side graph plots academics and financial institutions, the bottom right hand side graph plots employer organizations and trade unions. The grey line in (a) indicates CPI inflation in Norway. The medium-term horizon can be found in the Appendix Figure C.1.

question is how well different agents forecast inflation and whether their accuracy varies over time. Table 1 reports the forecast accuracy for the different groups of agents and forecast horizons measured as the root mean squared forecast error (RMSFE) and the relative RMSFE relative to the forecast errors of economists. The upper panel presents results for observations

prior to the recent inflationary episode, up to 2021Q2, whereas the lower panel covers the more recent period, from 2021Q3 to 2024Q4.¹⁰ During the period of low and stable inflation (2002Q1–2021Q2), households exhibit the least accurate forecasts, whereas economists provide the most accurate ones, both at the one- and two-year-ahead horizons. For the one-year horizon, social partners and business leaders perform equally well with no statistically significant differences in their accuracy compared to economists. For two-years-ahead forecasts instead, business leaders perform considerably worse than economists. The accuracy of all agents deteriorates substantially during the second sample period, but proportionally more for economists and social partners than for households. In fact, households perform best over the second sample, whereas the other three groups show similar accuracy. This shift reflects the fact that households had higher inflation expectations during the post-pandemic inflation period. However, the differences in accuracy are not statistically significant, most likely due to the short span of the high-inflation period. Because households constantly forecast high inflation, it is not surprising that their expectations are highly inaccurate in low inflation environments but become better aligned with actual inflation during high-inflation periods.

We now turn to the role of sociodemographic and firm characteristics for subjective inflation expectations. Table 2 presents summary statistics of 12-months-ahead inflation expectations by household groups for the period of low and high inflation, 2009Q1–2021Q2 and 2021Q3–2024Q4, respectively. Across both periods, women have higher inflation expectations than men (Jonung 1981, Bryan & Venkatu 2001, D’Acunto, Malmendier & Weber 2021, Reiche 2025), though the gender gap narrows during the high-inflation period. The relationship between age and inflation expectations also changes markedly across periods. During the low-inflation regime, expectations display a mild U-shape, with both younger and older households reporting higher expected inflation than middle-aged respondents. In contrast, during the high-inflation period, expectations decline monotonically with age: younger respondents report the highest expected inflation, whereas the oldest group reports the lowest. This shift might suggest that inflation salience and recent experience matter more for younger cohorts, who tend to update their expectations more strongly when inflation rises. Differences across labor-force status arise in both periods. Students and the unemployed, groups typically characterized by lower income and weaker attachment to the labor market, consistently expect higher inflation. Nonetheless, as inflation rose, expectations increased across all labor-force categories, narrowing the relative dispersion across groups. Finally, little systematic regional variation in household expectations is observed, consistent with evidence for the

¹⁰The sample refers to the corresponding forecast errors. Specifically, the second sample includes four-quarter-ahead forecasts made between 2020Q3 and 2024Q4, and eight-quarter-ahead forecasts made between 2019Q3 and 2023Q4.

Table 1: Accuracy of Inflation Forecasts

	Households	Business leaders	Social partners	Economists
Panel A: 2002Q1-2021Q2				
$h = 4$	1.50 1.34***	1.18 1.05	1.17 1.05	1.12
$h = 8$	2.22 1.83***	1.59 1.31***	1.24 1.02	1.21
Panel B: 2021Q3-2024Q4				
$h = 4$	1.78 0.88	2.10 1.04	2.06 1.01	2.03
$h = 8$	1.92 0.78	2.55 1.03	2.33 0.94*	2.47

Note: RMSFE (first row) and relative RMSFE (second row) of inflation expectations with respect to the economist groups. ‘hh’: households, ‘bl’: business leaders, ‘sp’: social partners, ‘ec’: all economists. ‘h’ denotes the forecasts horizon in quarters. ‘*’, ‘**’ and ‘***’ indicate significance levels of the Diebold and Mariano test of equal accuracy between each group with respect to the economist group, at the 10, 5 and 1 percent significance respectively.

United States (D’Acunto et al. 2023). Although mean expectations increased across all Norwegian regions during the high-inflation period, their relative ranking remained essentially unchanged.

For business leaders, Table 3 shows the level and dispersion of inflation expectations rose sharply in the high-inflation period. Average 12-month-ahead expectations increased from 2.7 percent in 2009Q1–2021Q2 to just above 7 percent in 2021Q3–2024Q4, with a much wider distribution of responses in the latter period. Sectoral heterogeneity is substantial and changes meaningfully across periods. During the low-inflation period, expectations were relatively similar across sectors. In the high-inflation period, however, dispersion widened markedly: business leaders of manufacturing firms reported the highest expectations, followed closely by firms in the construction sector. This shift indicates that inflation expectations among producers of goods and energy-intensive sectors rose more strongly once inflation pressures intensified. Larger firms have higher inflation expectations in the high-inflation period, whereas their expectations are similar to those of small firms in the low-inflation sample. Regional patterns also change across low- and high-inflation periods. Before 2021, inflation expectations were virtually identical across regions. By contrast, in the high-inflation period, business leaders in Oslo and South Norway reported the highest expectations. This

divergence likely reflects regional variation in energy prices after 2021Q3, as we discuss in Section 3.5.¹¹ Finally, business leaders' expectations are closely related to their own price assessments. Across both periods, business leaders anticipating higher sales or purchasing prices also report higher expected inflation. These relationships strengthen markedly in the high-inflation period, consistent with evidence that business leaders extrapolate from their own cost and pricing conditions (Andrade et al. 2022).

Group	2009 Q1 - 2021 Q2			2021 Q3 - 2024 Q4		
	Mean	SD	Observations	Mean	SD	Observations
All	3.02	5.07	50,609	4.93	7.9	14,070
Age						
< 35 years	3.37	6.52	10,886	5.49	9.11	4,181
35 to 54 years	2.79	4.42	17,904	5.02	7.83	4,728
55 to 64 years	2.91	4.53	9,005	4.60	7.07	2,234
≥ 65 years	3.17	4.95	12,814	4.27	6.70	2,927
Sex						
Man	2.79	4.91	35,510	4.90	7.99	8817
Woman	3.32	5.25	29,169	4.98	7.74	5253
Labor force						
Public_sector	2.94	4.64	19,511	4.81	6.87	3,016
Private_sector	2.69	4.89	22,931	4.96	7.96	6,236
Retired	3.26	5.55	20,683	4.66	7.48	3,285
Unemployed	3.75	9.08	9,659	5.46	11.43	179
Student	3.66	7.18	12464	5.56	10.00	1,227
Region						
North	3.04	5.03	11,905	5.07	8.45	3,303
South	3.03	5.07	32,048	4.97	7.83	8,239
Oslo	2.99	5.15	6,656	4.63	7.39	2,528

Table 2: Summary statistics of inflation expectations 12 months ahead by household group

The findings of this subsection largely confirm existing evidence on level differences in inflation expectations among different agents during periods of low and stable inflation. Specifically, we confirm that households tend to have the highest expectations and are the least accurate, whereas economists generally exhibit the lowest expectations and the highest accuracy. Business leaders fall in between, and social partners, our newly analyzed group,

¹¹Regional energy price differentials widened significantly after 2021, with southern regions experiencing much higher electricity costs than northern regions.

Group	2009 Q1 - 2021 Q2			2021 Q3 - 2024 Q4		
	Mean	SD	Observations	Mean	SD	Observations
All	2.68	2.99	28755	7.03	9.41	8164
Number of employees						
Less than 50	2.71	2.97	13703	6.69	8	2928
Greater than 50	2.74	3.28	12136	7.21	10.11	5235
Sector						
Manufacturing	2.58	2.58	7296	7.82	11.41	1763
Retail and trade	2.78	2.94	5443	6.57	7.72	1697
Building and construction	2.85	3.34	4023	7.66	10.06	1395
Services	2.66	3.13	11959	6.56	8.67	3312
Price Expectations						
↑ Sales	3.12	3.36	4095	8.78	11.41	2448
↓ or = Sales	2.6	2.91	24662	6.3	8.33	5716
↑ Purchasing	3.15	3.25	5139	8.85	11.61	3087
↓ or = Purchasing	2.57	2.91	23618	5.95	7.64	5077
Region						
North	2.69	2.58	5868	6.09	6.68	1640
South	2.67	3	17846	7.09	9.3	4477
Oslo	2.72	3.35	5050	7.62	11.23	2047

Table 3: Summary statistics of inflation expectations 12 months ahead by firm group

closely resemble economists in both levels and accuracy. However, our analysis also highlights notable differences during the recent period of high inflation: the rankings of accuracy among agents shift, with households performing better relative to experts. Additionally, disaggregating social partners reveals trade unions report systematically higher expectations during high-inflation periods, a pattern not observed in periods of low inflation (see Figure 1b). The same figure shows that economists have higher short-term expectations than professional forecasters, though the wedge is much larger in high-inflation periods. Regarding households, expectations increased proportionally more for men, respondents aged between 35 to 44 and those working in the private sector. Similarly, business leaders' expectations have become increasingly heterogeneous according to firm size, sector and region (Table 3). These findings highlight the importance of considering the role of varying inflation levels when interpreting survey results and comparing expectations across different agents.

Figure C.2 in the Appendix puts the cross-group dispersion in our survey into perspective by comparing our one-year-ahead inflation expectations with U.S. evidence from the

Michigan Survey of Consumers for households, the Survey of Firms’ Inflation Expectations (SoFIE) of the Cleveland Fed , and the Survey of Professional Forecasters of the Philadelphia Fed for professional forecasters. The U.S. data show larger household-professional gaps, although these differences are not substantially larger than in our survey. During the post-Covid inflation surge, households and firms revised their expectations more strongly than professional forecasters in both countries. However, firm expectations in the United States remained below household expectations on average, whereas business leaders’ expectations in our survey exceed those of households. Overall, this external benchmark reinforces the relevance of our multi-agent comparison: the broad ranking and post-Covid adjustment patterns are consistent with evidence from another country, while the more limited dispersion across Norwegian groups emerges as a distinctive feature of our survey. This feature could be an outcome of the harmonization of the survey, but it could also be consistent with Norwegian agents being relatively well informed. Our evidence does not allow us to distinguish between these explanations.

We replicate the summary statistics and accuracy comparison for wage expectations at a 12-month horizon in Appendix Tables [B.1](#) (Accuracy), [B.2](#) (Household summary statistics) and [B.3](#) (Business leaders summary statistics). The findings on forecasting accuracy are in line with those for inflation expectations. During the pre-pandemic period, both expert groups provide the most accurate forecasts, business leaders perform slightly worse, and households display the largest forecast errors. Differently from inflation, we do not observe the reversal in accuracy in the post-pandemic sample: on the contrary, the relative forecast accuracy of households and business leaders is unchanged, or even deteriorates.

The overall patterns of heterogeneity in wage expectations broadly mirror those observed for inflation expectations, although some differences emerge across periods and groups. Similar to inflation expectations, younger households and students tend to have higher wage growth expectations in both periods, possibly due to expected improvements in their employment situation. In contrast to inflation, the relationship between age and wage expectations remains monotonic in both pre- and post-inflation surge periods. Men tend to have higher wage expectations than women in both periods, and private-sector workers have higher wage expectations than workers in the public sector. The highest wage expectations for households by region are for Oslo, where most economic activity is concentrated.

Focusing on business leaders, we observe small differences in terms of size and sector of the firm for the low-inflation sample, whereas we observe somewhat higher expectations for larger firms and firms in the manufacturing sector for the high-inflation period. Business

leaders in the South and Oslo regions display the highest wage expectations, the latter in line with households. Interestingly, we don't observe differences in wage expectations based on business leaders' own sales or purchase prices in the low-inflation period, whereas in the high-inflation period, business leaders expecting an increase in own sales or purchasing prices also expect higher wages.

Overall, the comparison between inflation and wage expectations reveals the pass-through from inflation to wage-growth expectations might be incomplete and heterogeneous. Agents do not adjust one-for-one: even during the high-inflation period, expected wage growth rises by substantially less than expected inflation. This muted response suggests that nominal wage expectations remain only partially indexed to inflation. We analyze this inflation–wage expectations pass-through more rigorously in Section 3.4.

3.2 Cross-Sectional Disagreement

In addition to differences in the levels of inflation expectations, we are also interested in the dispersion of agents' expectations and their skewness, particularly because the right tail may provide predictive power for future inflation (Reis 2021).

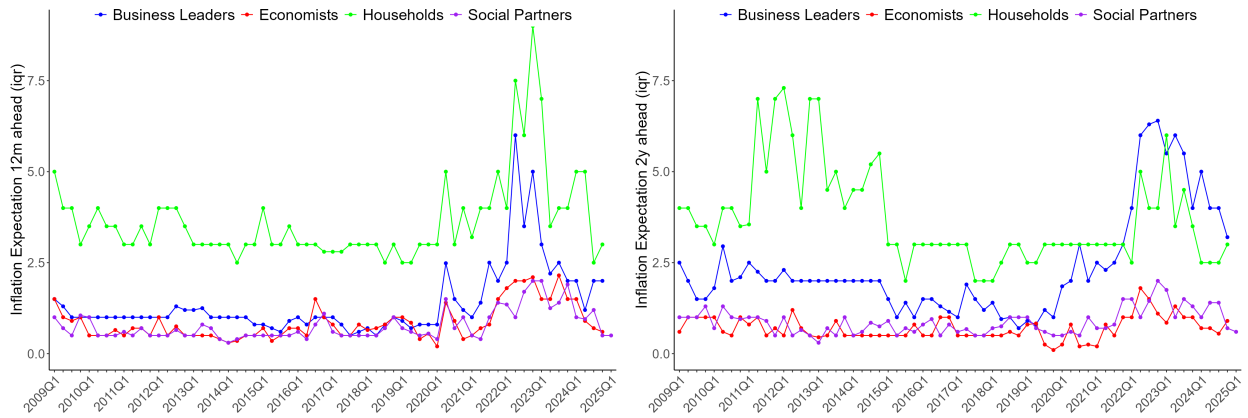


Figure 2: Timeseries of interquartile range of inflation expectations since 2009

Note: The left hand side graphs plot short-term expectations (over the next 12 months) and the right hand side graphs plot medium-term expectations (over the next 2 years).

Figure 2 displays the interquartile range of inflation expectations of households, business leaders, social partners and economists at short- and medium-term horizons. Households have the highest dispersion at the short horizon, more than twice as high as the dispersion of the other agents until the onset of the inflation surge, followed by business leaders. Economists and social partners show similarly low levels of disagreement throughout the

sample. With the onset of the rise of inflation, the dispersion among households and business leaders increases sharply, nearly doubling, whereas both social partners and economists display relatively little change in their disagreement about the inflation outlook. We observe similar patterns at the medium-term horizon until about 2021Q4 with households and business leaders having the largest disagreement in their inflation outlook. Starting in 2022, business leaders' dispersion increased more than the dispersion of households. Again, at the medium-term horizon, social partners and economists exhibit little disagreement even during the period of high inflation.

Examining skewness in inflation forecasts is one way to assess the risk of rising inflation due to a fat right tail of beliefs. We measure skewness as the difference between the average and median forecasts, with a positive value indicating a fat right tail and report it in Figure 3. Typically, household expectations exhibit the highest skewness, followed by those of business leaders, social partners, and economists. Notably, the latter two groups show virtually no skewness before the pandemic. However, after 2021, the mean forecasts of business leaders exceeded the median more than those of any other agent, indicating a pronounced right skew for both short-term and medium-term expectations. Households and social partners follow, whereas economists exhibit no significant change.

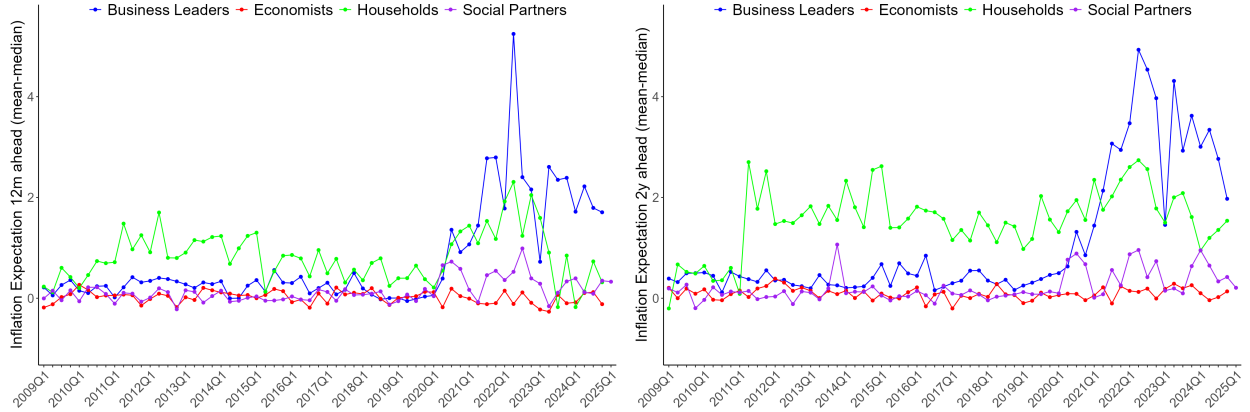


Figure 3: Timeseries of difference between mean and median inflation expectations since 2009

Note: The left hand side plots short-term expectations (over the next 12 months) and the right hand side plots medium-term expectations (over the next 2 years).

Compared to the other agents, both economists and social partners exhibit a mild and similar increase in dispersion during the post-pandemic inflation surge but the two groups differ sharply in their skewness. While economists observe an increase in dispersion and no change in skewness, social partners' distribution of inflation expectations becomes wider

and more right-skewed after 2020, suggesting the share of respondents who predicted high inflation increased somewhat more among social partners than among economists.

Similarly to level of inflation expectations, we observe a similar ranking for skewness and dispersion. During the sample period of low inflation, households exhibit the highest dispersion in inflation expectations, followed by business leaders, economists and social partners. Households also display substantial right skewness as do business leaders but to a lesser extent. The rankings substantially change after the pandemic. Business leaders' inflation expectations exhibit the highest degree of skewness and uncertainty at the two-year horizon, followed by households.

3.3 Term Structure

A central issue for central banks is the degree to which inflation expectations are anchored, which relates to trust in the institution and helps the bank to stir inflation towards the target. The inflation target of central banks is typically defined as an inflation number that the central bank aims to achieve over the medium term. Hence, one way to test whether inflation expectations are anchored is to test whether longer-term inflation expectations are centered at the inflation target of the central bank (level anchoring). Complementary, if inflation expectations are well anchored, it should be the case that short-term inflation expectations can fluctuate due to shocks hitting the economy, but agents should understand that the central bank will take the appropriate actions to bring inflation back to target so that variation in longer-term inflation expectations should be unresponsive to variation in short-term expectations (sensitivity anchoring) ([Weber et al. 2022](#), [D'Acunto et al. 2024](#)).

While previous research has studied the sensitivity of medium-term expectations to short-term expectations for households and firms, evidence for other agents and across inflation environments remains limited. In particular, little is known about how social partners, the actors directly involved in collective wage bargaining, form expectations across horizons, or how their behavior compares with that of other groups and across time. By combining harmonized data across these four agent types, we provide a comprehensive comparison of the pass-through from short- to medium-term inflation expectations, both in periods of low and stable inflation and during the recent inflation surge, allowing us to assess how expectations anchoring varies across different agents and inflation environments.

Figure 4 plots the inflation expectations of all four agents across different forecasting horizons. For the pre-pandemic sample, we find that, for both households and business leaders, medium-term inflation expectations are typically higher than 12-month-ahead ex-

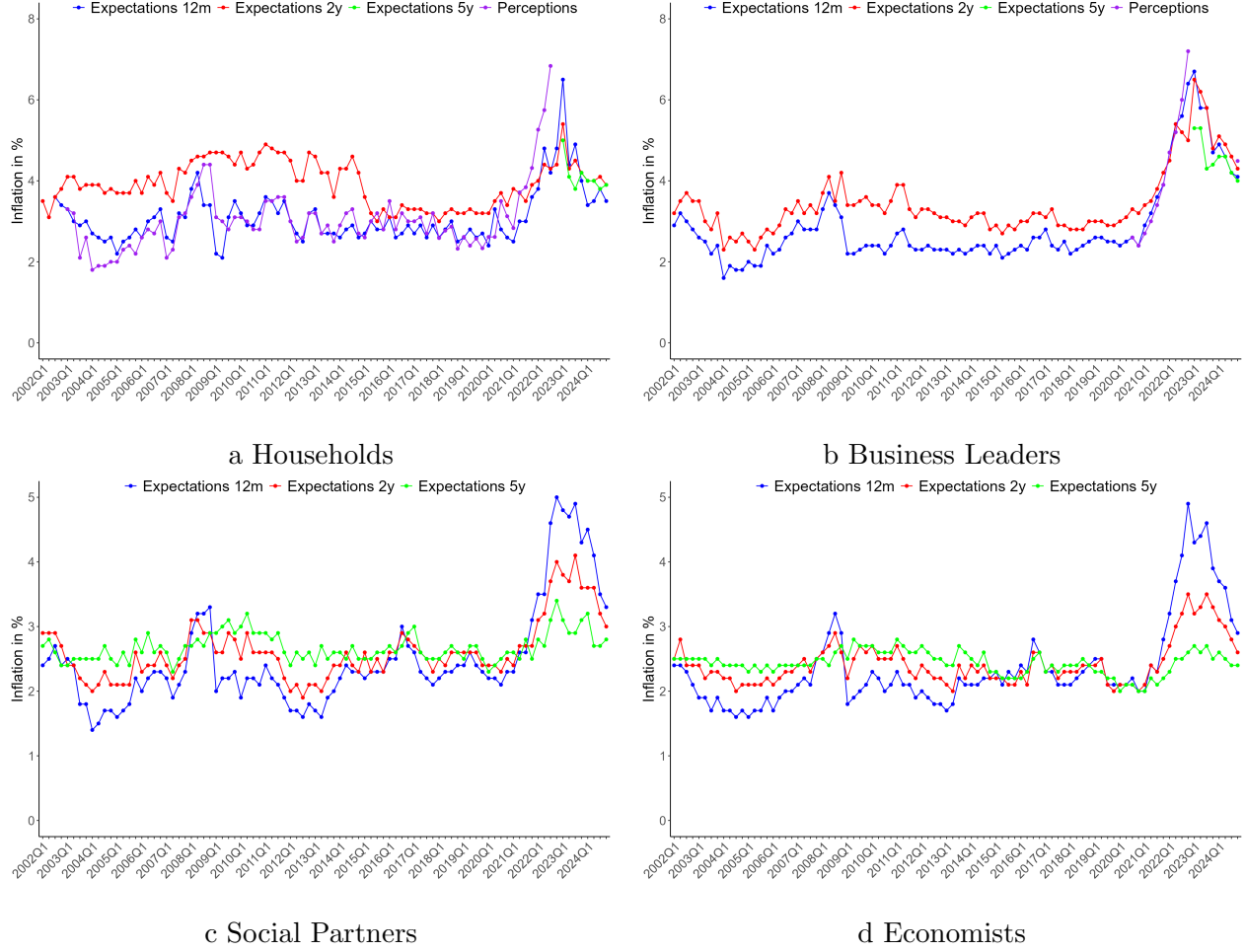


Figure 4: Time Series of Term Structure

pectations (or inflation perceptions for households), thus exhibiting an upward sloping term structure. For households, the result of an upward sloping term structure of inflation expectations resembles the findings from the New York Fed SCE (Armantier et al. 2013). The pass-through from inflation perceptions to short-term expectations is very high before the pandemic, but it declines substantially afterwards, when perceptions exceed one-year-ahead expectations. For all agents, the term structure of expectations shifts from sloping upward to downward after the pandemic. In general, for all groups, the longer the forecast horizon, the less volatile the inflation expectation. However, business leaders stand out in the high-inflation period: although their medium-term expectations remain below their short-term expectations, they increase at a similar pace. This pattern suggests that business leaders did not fully discount the inflation surge at longer horizons, a pattern consistent with the stronger pass-through from short- to medium-term expectations documented below. Again, social partners behave very similarly to economists. However, a notable difference lies in

their long-term inflation expectation. Whereas economists lowered their long-term forecasts following the announcement in March 2018 of the central bank to lower the inflation target from 2.5% to 2%, social partners did not revise their forecasts downward. This divergence might have had significant implications for wage-setting processes.

We next aim to understand how these aggregate results are related to the pass-through at the forecaster level, by estimating the degree to which short-term expectations drive medium-term inflation expectations. In line with the analysis in [Jonung \(1981\)](#) for households and [Candia, Weber, Gorodnichenko & Coibion \(2023\)](#) for firms, we estimate term structure regressions:

$$\mathbf{E}_{i,t}\pi_{t+8} = \alpha + \beta_1\mathbf{E}_{i,t}\pi_{t+4} + \beta_2\pi_t + \delta Y_t + \gamma X_i + \epsilon_{i,t} \quad (1)$$

where $\mathbf{E}_{i,t}\pi_{t+k}$ denotes the expectations of agent i of inflation over the next k quarters elicited at time t , π_t denotes realized inflation at time t over the previous 12 months, X_i is a vector of individual control variables. We also include other macro controls Y_t : the year-on-year growth rate of GDP and of the Brent oil price as well as the exchange rate (I44, in levels). Table 4 summarizes the pass-through of short-term inflation expectations into medium-term expectations. Panel A shows the results for the period of low and stable inflation and panel B for the post-pandemic inflation period. For sake of exposition, we report the full set of controls in Tables B.4-B.9 in the Appendix, together with alternative specifications of regression (1) which exclude macroeconomic controls, include time fixed effects or estimate the regression on the full sample. Our findings are robust to these alternative specifications.

Short-term expectations significantly influence medium-term forecasts for all agents, but the magnitude and dynamics of the pass-through vary sharply across groups and inflation regimes. In the low-inflation period, the pass-through is strongest for social partners (1.30) and business leaders (0.82), and lowest for households (0.70) and economists (0.71). During the high-inflation period, the gap widens: the pass-through substantially falls for households (0.44) and economists (0.56) but rises for business leaders (0.94), while it remains above one for social partners (1.20). These results indicate business leaders' medium-term expectations have become more sensitive to short-term expectations, whereas households and economists increasingly expect mean reversion. Social partners consistently exhibit an extrapolative component in their expectations, possibly reflecting their role in wage bargaining and reliance on recent inflation when forming medium-term views about inflation and wages. Taken together, the findings reveal substantial heterogeneity in expectations anchoring across groups of agents and across inflation regimes.

Although medium-term forecasts are less volatile than short-term ones, they respond strongly to revisions in short-term expectations. Because short-term expectations are shaped by perceptions (Huber et al. 2023), which themselves reflect recent inflation outcomes, agents appear to interpret inflation shocks as relatively persistent. The decline in pass-through for most groups during the high-inflation period suggests expectations of faster mean reversion and limited risks of de-anchoring. In contrast, business leaders’ rising pass-through indicates they perceive inflation shocks as more persistent, likely reflecting heightened sensitivity to input cost pressures and pricing dynamics. The evidence in Section 3.5 provides a concrete mechanism consistent with this interpretation. During the inflationary period, firms were directly exposed to the electricity price shock through their production costs, whereas households were largely insulated by a government support scheme. The regional divergence in electricity prices therefore created a shock that was especially salient for business leaders in the South. Later results show that business leaders exposed to higher electricity prices revised not only their short-term but also their two-year-ahead inflation expectations upward. The stronger response at the medium-term horizon suggests that firms did not interpret the shock merely as a temporary relative-price change, but as a source of more persistent inflationary pressure. This interpretation is also consistent with the increased pass-through from inflation expectations to wage expectations below. Taken together, these results suggest that business leaders’ weaker sensitivity anchoring is closely related to their direct exposure to cost shocks and to their role in price and wage setting.

For social partners, a persistently high pass-through might reflect the salience of recent inflation in wage negotiations. Differences in perceived persistence are economically significant because they influence how expectations feed into price and wage setting. When business leaders and social partners expect inflation shocks to last, they are more likely to preemptively raise prices or negotiate higher wages, potentially amplifying inflation dynamics (Abberger et al. 2024). Conversely, households’ and economists’ stronger belief in mean reversion hints towards more anchored expectations, which possibly support monetary policy credibility and helps stabilize inflation outcomes. Recognizing this heterogeneity is therefore crucial for central banks aiming to assess anchoring, to communicate effectively, and to calibrate policy responses across different economic agents.

3.4 Pass-through of Inflation Expectations to Wage Expectations

Having established how short-term inflation expectations shape medium-term expectations across agents, we next examine how these expectations feed into wage growth forecasts. The NBES provides a unique setup to study this wage–inflation pass-through in expectations,

Table 4: Term Structure at the Individual Level

	$E_{i,t}\pi_{t+8}$			
	Households	Business Leaders	Economists	Social Partners
Panel A: 2009 Q1 - 2021 Q2				
$E_{i,t}\pi_{t+4}$	0.695*** (0.006)	0.824*** (0.007)	0.710*** (0.016)	1.297*** (0.023)
Micro Controls	Yes	Yes	Yes	Yes
Macro Controls	Yes	Yes	Yes	Yes
Observations	30,008	18,452	2,918	1,563
R ²	0.311	0.467	0.463	0.675
Panel B: 2021 Q3 - 2024 Q4				
$E_{i,t}\pi_{t+4}$	0.442*** (0.008)	0.941*** (0.008)	0.556*** (0.022)	1.200*** (0.028)
Micro Controls	Yes	Yes	Yes	Yes
Macro Controls	Yes	Yes	Yes	Yes
Observations	10,353	6,481	603	489
R ²	0.246	0.687	0.656	0.800

*p<0.1; **p<0.05; ***p<0.01

Note: Regression from model (1) for households, business leaders, economists and social partners. All models include individual level controls. For households, controls include age, sex and occupational status; for business leaders sector and size; for economists a dummy for employment at financial institution and for social partners a dummy for employer organizations. Macro controls include CPI inflation (YoY), GDP growth (YoY), the exchange rate (I44 in levels) and the annual change in the Brent oil price - all contemporaneously.

because it collects expected wage growth for all groups, in addition to inflation expectations. Households are asked about their personal wage growth over the past 12 months and their expected personal wage growth over the same horizon. For business leaders, economists, and social partners, the survey focuses on expected wage growth for the current and next year. Business leaders are specifically asked about wage growth in their own business, whereas the other two groups are asked about expected aggregate wage growth.

Social partners significantly affect actual wage growth, as they set the wage norm for each sector, which in turn forms the basis for individual wage negotiations. This role of social partners in shaping actual wage growth reflects the broader institutional framework of wage setting in Norway. While wage negotiations are formally decentralized, they are coordinated through the so-called *frontfagsmodellen*, in which the manufacturing sector negotiates first and sets a wage norm that guides settlements in other sectors. The rationale for this model is that the wage settlements should ensure competitiveness of the trading sector, that is, the manufacturing industries exposed to international competition. This coordination gives social partners an important influence in anchoring wage growth across the economy, making their expectations especially informative. In terms of collective bargaining coverage, Norway reaches about 70%, which places it broadly in line with several European countries.¹² Unlike many other European countries, Norway achieves this coverage primarily through voluntary coordination rather than legal extension of agreements. Thus, while the Norwegian model is distinct in its reliance on a negotiated wage norm and voluntary diffusion, it is not an outlier in terms of overall bargaining coverage. This institutional context helps explain why wage expectations, particularly those of social partners and business leaders, are closely linked to actual wage outcomes (see Table B.1 in the Appendix), and why they provide a valuable perspective on the wage-inflation pass-through.

We first plot the time series of nominal wage growth and inflation expectations over a 12-month horizon in Figure 5. Between 2002 and 2008, all agents expected nominal wage growth to exceed inflation. Following the Great Financial Crisis, households' wage growth expectations fell markedly and remained below inflation expectations for most of the subsequent period. In contrast, business leaders, social partners, and economists continued to expect real wage gains until the onset of the Covid-19 pandemic. Since then, business leaders, like households, anticipated a decline in real wages, whereas social partners and economists expected wage growth and inflation to move more closely together. These patterns suggest the expected wage-inflation pass-through might differ across agents and respond to changing

¹²The coverage is somewhat lower than in the other Nordic countries (Sweden, Denmark, and Finland) where coverage typically exceeds 85%, but it is similar to or slightly below levels in Italy, France, Spain, and the Netherlands (ranging from 75–90%), and higher than in Germany, where coverage is closer to 45%.

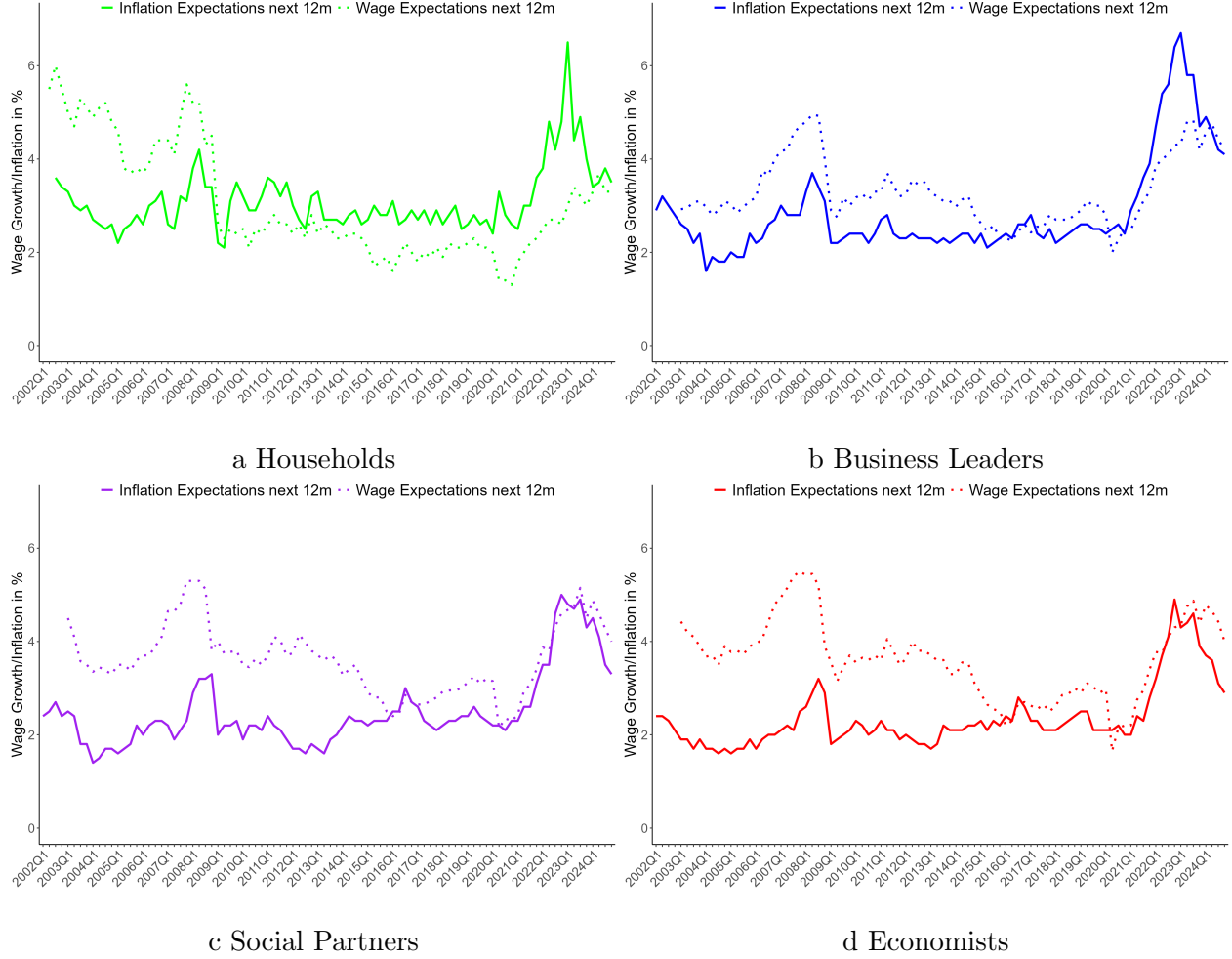


Figure 5: Timeseries of Wage and Inflation Expectations (12m ahead)

Note: For business leaders, unions and experts the 12 months ahead forecast is computed as a weighted average of current year and next year forecast where the weights are determined by the current quarter. Business leaders and households forecast individual wage growth, unions and experts wage growth in the aggregate economy.

economic conditions. This time variation may have important implications for price-wage dynamics, complementing recent work focusing mainly on the post-2020 period (Hajdini et al. 2022, Savignac et al. 2024, Abberger et al. 2024, Baumann et al. 2024, Buchheim et al. 2024).

To analyze how inflation expectations shape wage expectations more formally, we follow Jain et al. (2024) and estimate the following regression:

$$\mathbf{E}_{i,t}w_{i,t+4} = \alpha_0 + \beta_1\mathbf{E}_{i,t}w_{i,t} + \beta_2\mathbf{E}_{i,t}\pi_{t+4} + \delta Y_t + \gamma X_i + \epsilon_{i,t}, \quad (2)$$

where $\mathbf{E}_{i,t}w_{i,t+k}$ is agent i 's survey response for the expectation about (personal) wage growth over the next k quarters and $\mathbf{E}_{i,t}w_{i,t}$ is agent i 's perceived (personal) wage growth over the past

12 months (capturing persistence in wage growth expectations). Because the question is asked in terms of current and next calendar year for business leaders, economists and social partners, we construct the fixed horizon forecast for the next twelve months as a weighted average of the forecasts for the current and next year, with weights given by the proportion in forecasting horizon, following [Dovern et al. \(2012\)](#). To approximate wage growth perceptions of business leaders and experts, we use a weighted average of individual current year expectations and lagged nominal wages of the past year that are publicly available. We include the same set of macroeconomic and individual controls as for the term structure regression.

Table 5 reports the results. For sake of exposition, we report the coefficients on inflation expectations and nominal wages (perceptions/lagged wages) only.¹³ We observe a positive and significant pass-through of inflation to wage expectations for all agents but the magnitude differs across types and time periods. In the pre-pandemic sample, business leaders exhibit the highest pass-through (0.31), followed by economists (0.22), social partners (0.12) and households (0.06). After the inflation surge, the pass-through is higher for business leaders (0.33), economists (0.23), and social partners (0.19), but lower for households (0.05), who have the lowest pass-through of all agents. Overall, despite this heterogeneity, the pass-through is incomplete, in that there is less than a one-for-one relationship for all agents and both sub-samples.

We compute a similar regression as (2) using realized data, to estimate the effective pass-through from inflation to wages and to benchmark the subjective expectations results. We estimate a coefficient associated with lagged wage growth of 0.7 and one associated with either contemporaneous or lagged inflation of about 0.17 and 0.12, respectively, and non-statistically significant (see Table B.22). Therefore, in most instances, agents fail to fully recognize the persistence of wage growth and overestimate the role of (future) inflation in determining future wage growth.

Several recent studies have estimated the pass-through of inflation expectations to households’ and business leaders’ wage expectations, mostly using randomized controlled trials during the high-inflation period. Our estimated pass-through for households is in line with the findings in the literature, which range between 0 and 0.2 ([Jain et al. 2024](#), [Buchheim et al. 2024](#), [Hajdini et al. 2022](#)). We also confirm the estimates of the inflation-wage pass-

¹³We report the coefficients for the full set of controls in Tables B.10-B.17 in the Appendix, where we also report alternative specifications of regression (2) which exclude macroeconomic controls, include time fixed effects, use only Q1 responses and exclude wage growth perceptions. Because wage expectations are reported for a fixed horizon (the current and the following year), focusing on Q1 respondents ensures that ‘next year’ uniformly corresponds to the first quarter of the subsequent year. Tables B.18-B.21 in the Appendix provide estimates for the full sample. Overall, our results are robust across alternative specifications.

Table 5: Inflation-Wage Expectations Pass-through

	$E_{i,t}w_{t+4}$			
	Households	Business Leaders	Economists	Social Partners
Panel A: 2009 Q1 - 2021 Q2				
$E_{i,t}w_t$	0.228*** (0.007)	0.354*** (0.005)	0.081*** (0.007)	0.046*** (0.009)
$E_{i,t}\pi_{t+4}$	0.062*** (0.015)	0.305*** (0.005)	0.221*** (0.020)	0.116*** (0.019)
Micro Controls	Yes	Yes	Yes	Yes
Macro Controls	Yes	Yes	Yes	Yes
Observations	11,906	18,629	2,911	1,551
R ²	0.102	0.448	0.389	0.359
Panel B: 2021 Q3 - 2024 Q1				
$E_{i,t}w_t$	0.067*** (0.008)	0.761*** (0.010)	-0.007 (0.010)	0.057*** (0.016)
$E_{i,t}\pi_{t+4}$	0.052*** (0.013)	0.329*** (0.007)	0.233*** (0.026)	0.189*** (0.022)
Micro Controls	Yes	Yes	Yes	Yes
Macro Controls	Yes	Yes	Yes	Yes
Observations	7,058	6,203	610	485
R ²	0.033	0.738	0.378	0.310

*p<0.1; **p<0.05; ***p<0.01

Note: Regression from model (2) for households, business leaders, economists and social partners. All models include individual level controls. For households, controls include age, sex and occupational status; for business leaders sector and size; for economists a dummy for employment at financial institution and for social partners a dummy for employer organizations. Macro controls include CPI inflation (YoY), GDP growth (YoY), the exchange rate (I44 in levels) and the annual change in the Brent oil price - all contemporaneously.

through for firms in Coibion et al. (2018), Savignac et al. (2024), Baumann et al. (2024), Abberger et al. (2024), Buchheim et al. (2024), Gautier et al. (2025), which report point estimates between 0.1 and 0.3. Yet, so far no estimates have existed for the pass-through among social partners and economists, nor for the increase in pass-through for these agents during the recent surge in inflation.

3.5 The Pass-Through of Electricity Prices

In 2022, the Norwegian economy, like most European countries, experienced a shock to electricity prices. However, what set Norway apart was that this shock was confined to the southern regions (Vestlandet, Østlandet, Sørlandet, Oslo), leaving the northern regions (Nord Norge and Midt Norge) largely unaffected. The southern price spike was exogenous to domestic macroeconomic conditions, but reflected the interaction of external demand pressures and local supply constraints. On the demand side, high electricity prices reflected a substantial increase in electricity exports of southern Norway towards Europe due to the energy crisis. On the supply side, record low levels of water reservoirs in southern Norway throughout most of the year restricted hydroelectric production. Given that 96% of Norway’s electricity supply is hydro-based and the fact that transmission capacity between the northern and southern regions is severely limited due to insufficient infrastructure, the supply shortages in the south put significant upward pressure on electricity prices.

As a result, electricity prices in Norway began diverging sharply between the two regions starting in 2021Q2. Before, prices across the two regions were virtually identical, see Figure 6a. Only firms, however, faced this divergence: starting in 2021Q4, the government introduced a household support scheme that largely offset the price increase in the south.¹⁴ The regional price divergence creates a close to ideal framework for evaluating the effects of electricity prices on business leaders’ expectations. By exploiting the geographical granularity of the NBES survey, we study the impact of energy prices on business leaders’ and households’ inflation expectations.

We estimate the pass-through of electricity prices to inflation expectations for business leaders using the following single difference regression in the spirit of Jo & Klopach (2025) and Wehrhofer (2023):

$$\mathbf{E}_{i,r,s,t}\pi_{t+k} = \alpha_0 + \beta D_{r,t} + \gamma_1 X_i + \gamma_2 T_t + \gamma_3 S_i + \epsilon_{i,r,s,t} \quad (3)$$

¹⁴Nearly all household electricity contracts (98%) are spot-priced, whereas firms’ exposure varies by sector: 94% in services, 87% in non-energy-intensive manufacturing, and 100% in energy-intensive manufacturing.

where $\mathbf{E}_{i,r,s,t}\pi_{t+k}$ denotes business leader i 's inflation expectations k quarters ahead elicited at time t , residing in part of the country r and belonging to sector s , $D_{r,t}$ is a dummy variable which takes the value of one during the periods in which the South region experiences higher electricity prices relative to the North (2021:Q2 until 2024:Q4). The dummy captures the effect of experiencing different energy prices on agents' expectations. We also include industry and firm size controls for business leaders, demographic controls for households, quarter-time fixed-effects, T_t , which account for changes in domestic or international macroeconomic conditions, such as monetary policy changes, fiscal policy interventions, shocks to commodity prices or supply chain disruptions, as well as a dummy for residing in the South, S_i , to capture constant differences across the two regions.

The identification strategy relies on two key assumptions. First, in the absence of the electricity price shock, business leaders' inflation expectations in the South and North would have followed parallel trends. While this assumption is inherently not testable, we can verify that before the shock, inflation expectations for business leaders across the two regions were on similar trends. Our identification assumption then boils down to assuming that absent the shock, the parallel trends had continued. Figure 6b shows mean expectations in both regions evolved nearly identically prior to 2021Q2 for both business leaders and households, supporting the plausibility of parallel trends. Second, the price increase in the southern region was exogenous to Norwegian macroeconomic conditions. Electricity prices rose due to external demand pressures from Europe's energy crisis combined with region-specific hydrological constraints, not because of domestic economic developments.

The parameter of interest in regression (3) is the coefficient associated with the shock dummy, β , which measures the average difference in inflation expectations between the North and the South due to the electricity price shock during the period for which electricity prices diverged. The first column of Table 6 presents the regression results for business leaders' overall inflation expectations for the whole Norwegian economy over the next 12 months, whereas the second column reports the results for expectations over the next two years. The estimates reveal a significant positive effect of $D_{r,t}$ on the inflation expectations of business leaders. On average, during the quarters when electricity prices diverged (2021Q2–2024Q4), business leaders in the South expected approximately 1.05 to 1.3 percentage points higher inflation than business leaders in the North. The stronger response for two-year-ahead inflation expectations to a temporary electricity shock is striking. Business leaders appear to perceive the electricity price shock to result in persistent inflationary pressures. This finding aligns with the earlier evidence on a large pass-through from short- to medium-term inflation expectation for business leaders during the inflationary surge: both patterns indicate that

business leaders perceive inflation shocks as persistent and adjust their medium-term views accordingly.

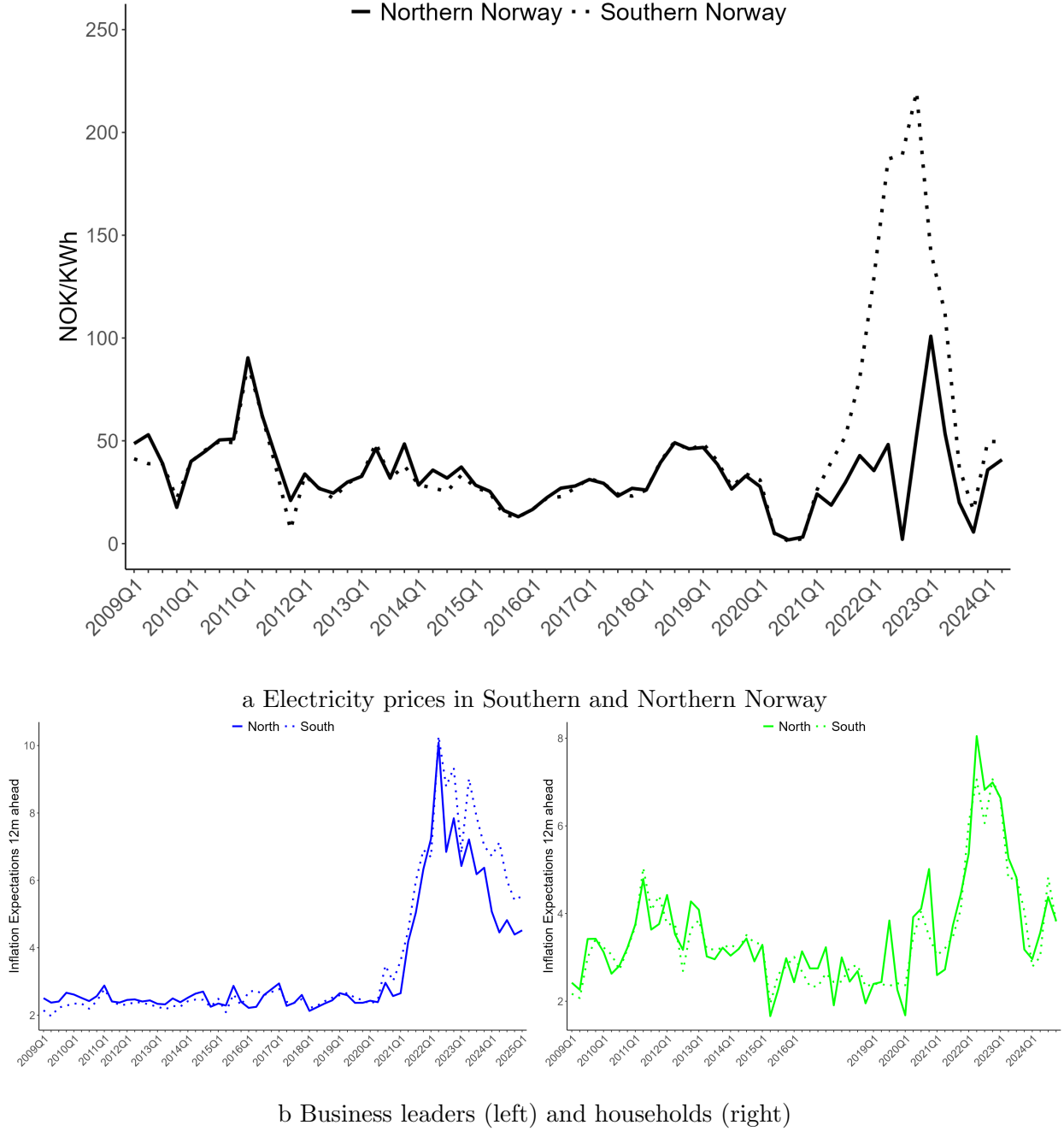


Figure 6: Timeseries of electricity prices and average inflation expectations split by region (12 months ahead) since 2009

We next compare these results to households. Figure 6 shows household inflation expectations in the North and South evolved almost identically throughout the sample period. To confirm this formally, we re-estimate equation (3) using household data. Table 6 shows

the regional divergence in electricity prices did not translate into a corresponding divergence in household inflation expectations. This null result is unsurprising given the government’s compensation scheme, which largely offset the increase in electricity costs for households in the South. Consequently, households across regions faced similar effective electricity prices. In the Appendix (Table B.23), we interact the treatment dummy with the log price differential between regions and find consistent results: a significant positive effect on business leaders’ inflation expectations, but none for households.

As a robustness check, we conduct a placebo test using a moving-window approach to assess whether the estimated treatment effects are specific to the actual intervention period. Specifically, we re-estimate our main regression model using placebo treatment dummies defined over all possible 3-year windows across the sample period. In each iteration, we define a placebo dummy variable equal to one for observations in the “South” region during the respective window and zero otherwise. We then regress the outcome on this placebo dummy, along with the full set of controls used in the main specification (3). Appendix Figure C.4 plots the estimated coefficients and their 95% confidence intervals for each window. We assess whether significant effects are detected outside the actual treatment period. The results show no consistent or systematic effects prior to the intervention, supporting the validity of the causal interpretation of our main findings.

Recent papers by Patzelt & Reis (2024) and Wehrhofer (2023) show a significant and positive pass-through from electricity prices to inflation expectations. In our case, we see a generalized increase in inflation expectations when electricity prices increase. However, expectations of households located in the South do not respond differently than those of households located in the North, because households were not exposed to different electricity prices (in contrast to the cited studies above). Regarding firms, Wehrhofer (2023) finds a positive and significant pass-through from electricity prices to own output prices, but documents that firms do not seem to extrapolate from individual prices to inflation for the overall economy. The divergence with respect to our results might be due to the fact that in Germany electricity contracts are usually fixed contracts, whereas in Norway, the overwhelming majority is based on spot prices. Therefore, firms in Norway might anticipate that all other firms will be exposed to the same increase in input cost and be forced to raise sales prices, which will ultimately results in a generalized increase in inflation.

Table 6: Effect of Energy Prices on Inflation Expectations

	Firms		Households	
	$E_{i,r,s,t}\pi_{t+4}$	$E_{i,r,s,t}\pi_{t+8}$	$E_{i,r,s,t}\pi_{t+4}$	$E_{i,r,s,t}\pi_{t+8}$
$D_{r,t}$	1.051*** (0.131)	1.287*** (0.155)	-0.067 (0.113)	0.0005 (0.151)
Time FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	45,521	44,079	83,031	73,323
R ²	0.176	0.130	0.045	0.037

*p<0.1; **p<0.05; ***p<0.01

Note: Regression from model (3) for business leaders and households. All models include time fixed effects, a dummy for living in the south and individual-level controls. For households, controls include age, sex and occupational status, while sector and size for business leaders.

4 Conclusion

We document new evidence on heterogeneity in inflation expectations across households, business leaders, social partners, and economists, using harmonized Norwegian survey data spanning the pre- and post-inflation periods. Several features make the survey ideal for a cross-agent comparison: first, the time span is relatively long compared to alternative surveys in the literature, as it has been running since 2002 at the quarterly frequency. Second, it surveys social partners, who are the key players in the determination of wages. Third, it elicits expectations about inflation and wages at multiple horizons. Fourth, the information sets available to agents when making their forecasts are directly comparable across groups and individuals, as respondents are surveyed in the same period, which does not include monetary policy decisions nor releases of national accounts. Finally, expectations are elicited in homogeneous ways ensuring that different wordings of questions can not drive heterogeneity across agents. These features allow us to study the evolution of expectations for both inflation and wages over the business cycle and in response to large shocks, and to compare it among different agents.

Consistent with other studies, we find that until 2021, households expectations were higher than business leaders, and economists (professional forecasters) had the lowest expectations. As a novel result, we document that social partners behave very similarly to economists. After the inflation surge, this ranking reversed: business leaders' expectations rose above those of households and remained elevated through 2024, reflecting both higher exposure to energy-price shocks and changing perceptions of inflation persistence.

Disagreement in short-term expectations was stable before the pandemic but rose sharply during the inflation surge, especially for households and business leaders. Across most of the sample, disagreement was largest for households, followed by business leaders, and lowest for social partners and economists. However, after 2021, business leaders' disagreement increased markedly and temporarily exceeded that of households, reflecting heightened uncertainty. Social partners and economists consistently displayed low and stable dispersion.

Turning to the term structure of expectations, short-term expectations play a central role in shaping medium-term expectations for all groups. During the low-inflation period, this short-to-medium-term pass-through was strongest for social partners (above 1), followed by business leaders, households, and economists. After 2021, the pass-through increased notably for business leaders, suggesting they perceived inflation as more persistent, but declined for households and economists, who anticipated faster mean reversion. This heterogeneity in perceived persistence helps explain how different groups adjusted to the new inflation regime and has potential implications for the speed at which shocks feed into prices and wages.

Regarding the relationship between wage and inflation expectations, we find expected inflation positively affects expected wages for all groups, though its magnitude remains well below one. The strength of this link differs across agents and over time: it is highest for business leaders, followed by economists and social partners, and it is lowest for households. During the recent inflation surge, the pass-through increased for business leaders, social partners, and economists, whereas it decreased for households.

Finally, we exploit a regional electricity-price shock to study its causal effect on inflation expectations. Business leaders exposed to higher energy prices revised their inflation expectations upward, especially for the two-year horizon, suggesting they viewed the shock as somewhat persistent. Households, in contrast, exhibited no significant response, reflecting the mitigating effects of government energy-support measures.

Taken together, these results highlight that the formation of inflation expectations differs systematically across agents and inflation environments. The heterogeneity we document, across households, business leaders, social partners, and economists, varies with the broader inflation regime, suggesting potential implications for inflation dynamics and the transmission of monetary policy.

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A Survey Questionnaire

Survey questions for inflation and wage expectations for households (hh), business leaders (bl), social partners (sp) and economists (ec).

Perceptions

- [hh] Compared to 12 months ago, do you think the prices of goods and services, as measured by the consumer price index, are higher, about the same or lower now?
- [hh] Approximately how much do you think prices have changed in percentage terms? (average)
- [bl] What do you think the general rise in prices for goods and services has been the last 12 months, as measured by the 12-month change in the consumer price index (CPI)?

Expectations

- [hh] Do you think that the prices of goods and services, as measured by the consumer price index, will be higher, unchanged or lower than today over the next 12 months?
- [hh] Approximately how much higher/lower, measured in percentage?
- [bl, sp, ec] What do you think the general rise in prices for goods and services will be in 12 months, as measured by the 12-month change in the consumer price index (CPI)?
- [hh] How much do you think the prices of goods and services, measured by the consumer price index, will increase annually in 2-3 years, measured in percentage?
- [bl, sp, ec] What do you think the general rise in prices for goods and services will be in two years, as measured by the 12-month change in the consumer price index (CPI)?
- [hh, bl, sp, ec] What do you think the general price increase on goods and services is in 5 years, as measured by the 12-month change in the consumer price index (CPI)?

Purchasing/Sale Prices

- [bl] Do you expect that over the next 12 months your business's purchase prices will increase more than, increase at approximately the same pace as or increase less than in the past 12-month period?
- [bl] Do you expect that over the next 12 months your business's selling prices will increase more than, increase at approximately the same pace as or increase less than in the past 12-month period?

Wages

- [hh] Compared with 12 months ago, by how much do you think your wages or pension has changed in percent?
- [bl] What do you expect the annual wage growth in your company will be this current year?
- [sp, ec] What do you think the average yearly growth in wages will be this current year?
- [hh] By approximately how much do you think your wages or pension will change in percent over the next 12 months? - [bl] What do you expect the average annual wage growth in your company will be next year?
- [sp, ec] What do you think the average yearly growth in wages will be next year?

B Tables

Table B.1: Accuracy of Wage Forecasts

	<i>households</i>	<i>business leaders</i>	<i>social partners</i>	<i>economists</i>
2002Q1-2021Q2				
<i>current</i>	0.75	0.57	0.41	0.44
<i>year</i>	1.70***	1.28	0.93	
<i>next</i>	1.09	0.61	0.60	0.59
<i>year</i>	1.84***	1.02	1.02	
2021Q3-2024Q4				
<i>current</i>	1.70	0.58	0.44	0.43
<i>year</i>	3.98***	1.35***	1.03	
<i>next</i>	2.14	1.15	1.09	1.23
<i>year</i>	1.73***	0.93**	0.88***	

Note: RMSFE for wage growth expectations. Households report perceptions over the past twelve months (cy) and expectations over the next twelve month (ny) of their own wage. Business leaders, social partners and economists are asked about fixed date forecasts: current year or next year wage. Households and business leaders are asked about their own wage bills, while social partners and economists are asked about aggregate wages.

Group	2009 Q1 - 2021 Q2			2021 Q3 - 2024 Q4		
	Mean	SD	Observations	Mean	SD	Observations
All	3.06	8.25	50609	4.43	10.2	14070
Age						
< 35 years	5.68	14.7	10886	7.07	16.03	4181
35 to 54 years	3.02	6.57	17904	4.01	7.12	4728
55 to 64 years	2.33	5.44	9005	3.47	6.85	2234
≥ 65 years	1.68	3.71	12814	2.37	4.43	2927
Sex						
Man	3.43	9.03	26693	4.76	10.39	8817
Woman	2.61	7.2	23916	3.85	9.82	5253
Labor force						
Public_sector	2.74	6.1	19511	3.46	6.83	3016
Private_sector	3.69	8.97	22931	4.88	9.35	6236
Retired	1.67	3.79	20683	2.37	3.68	3285
Unemployed	7.02	19.89	9659	17	32.24	179
Student	8.19	20.79	12464	9.7	22.37	1227
Region						
North	2.93	7.98	11905	4.41	10.55	3303
South	3.03	8.05	32048	4.25	9.68	8239
Oslo	3.39	9.53	6656	5.05	11.29	2528

Table B.2: Summary statistics of wage expectations by household group

Group	2009 Q1 - 2021 Q2			2021 Q3 - 2024 Q4		
	Mean	SD	Observations	Mean	SD	Observations
All	3.1	2.7	28755	5.41	8.16	8164
Number of employees						
Less than 50	3.11	2.62	13703	4.98	6.7	2928
Greater than 50	3.05	2.99	12136	5.65	8.88	5235
Sector						
Manufacturing	3.03	2.57	7296	6.02	9.92	1763
Retail and trade	3.03	2.64	5443	4.93	7.04	1697
Building and construction	3.18	3.17	4023	5.37	7.57	1395
Services	3.14	2.62	11959	5.34	7.87	3312
Price Expectations						
↑ Sales	3.35	3.63	4095	6.46	9.64	2448
↓ or = Sales	3.05	2.47	24662	4.97	7.42	5716
↑ Purchasing	3.18	3.3	5139	6.47	10.36	3087
↓ or = Purchasing	3.08	2.52	23618	4.79	6.46	5077
Region						
North	3.09	2.15	5868	4.78	6.15	1640
South	3.09	2.75	17846	5.17	7.41	4477
Oslo	3.12	3.01	5050	6.4	10.61	2047

Table B.3: Summary statistics of own wage expectations next year by firm group

Table B.4: Term Structure on the Individual Level — 2009 Q1–2021 Q2

	$E_{i,t}\pi_{t+8}$					
	Households			Business Leaders		
π_t	−0.359*** (0.040)	−0.118*** (0.044)		0.029 (0.024)	0.052** (0.026)	
$E_{i,t}\pi_{t+4}$	0.700*** (0.006)	0.695*** (0.006)	0.691*** (0.006)	0.823*** (0.007)	0.824*** (0.007)	0.787*** (0.007)
Female	0.664*** (0.071)	0.649*** (0.071)	0.675*** (0.071)			
Age	−0.015*** (0.002)	−0.015*** (0.002)	−0.016*** (0.002)			
Private_sector	−3.099*** (0.828)	−2.874*** (0.824)	−2.882*** (0.823)			
Public_sector	−2.931*** (0.829)	−2.785*** (0.825)	−2.791*** (0.824)			
Retired	−2.500*** (0.830)	−2.327*** (0.826)	−2.323*** (0.824)			
Student	−0.892 (0.836)	−0.634 (0.832)	−0.611 (0.831)			
Unemployed	−1.876** (0.875)	−1.692* (0.870)	−1.721** (0.869)			
Manufacturing				−0.244*** (0.066)	−0.252*** (0.066)	−0.228*** (0.066)
Retail_trade				−0.177** (0.071)	−0.178** (0.071)	−0.159** (0.071)
Services				−0.345*** (0.061)	−0.337*** (0.061)	−0.296*** (0.062)
Size > 50				−0.229*** (0.041)	−0.212*** (0.041)	−0.128*** (0.047)
GDP		0.026 (0.029)			−0.017 (0.015)	
I44		−0.071*** (0.004)			−0.009*** (0.002)	
Oil		−0.002** (0.001)			0.001 (0.001)	
Constant	6.863*** (0.834)	13.310*** (0.927)	7.319*** (0.854)	1.559*** (0.075)	2.395*** (0.241)	1.937*** (0.139)
Time FE	No	No	Yes	No	No	Yes
Observations	30,008	30,008	30,008	18,452	18,452	18,452
R ²	0.303	0.311	0.315	0.467	0.467	0.475

*p<0.1; **p<0.05; ***p<0.01

Table B.5: Term Structure on the Individual Level — 2021 Q3–2024 Q4

	$\mathbf{E}_{i,t}\pi_{t+8}$					
	Households			Business Leaders		
π_t	0.121*** (0.043)	0.071 (0.074)		−0.003 (0.051)	0.037 (0.086)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.443*** (0.008)	0.442*** (0.008)	0.445*** (0.008)	0.941*** (0.008)	0.941*** (0.008)	0.936*** (0.008)
Female	0.415*** (0.134)	0.416*** (0.134)	0.413*** (0.134)			
Age	−0.050*** (0.005)	−0.050*** (0.005)	−0.050*** (0.005)			
Private_sector	−3.528*** (1.188)	−3.529*** (1.188)	−3.511*** (1.189)			
Public_sector	−3.250*** (1.192)	−3.246*** (1.192)	−3.234*** (1.193)			
Retired	−1.858 (1.199)	−1.858 (1.199)	−1.846 (1.199)			
Student	−2.518** (1.210)	−2.507** (1.210)	−2.486** (1.210)			
Unemployed	−2.344* (1.320)	−2.332* (1.321)	−2.265* (1.321)			
Manufacturing				−0.621*** (0.241)	−0.621*** (0.241)	−0.672*** (0.230)
Retail_trade				−0.582** (0.241)	−0.577** (0.241)	−0.609*** (0.230)
Services				−0.682*** (0.213)	−0.672*** (0.213)	−0.716*** (0.204)
Size > 50				−0.117 (0.156)	−0.117 (0.156)	−0.096 (0.149)
GDP		0.052 (0.104)			0.236* (0.123)	
I44		−0.035 (0.045)			0.054 (0.053)	
Oil		−0.008 (0.008)			0.002 (0.009)	
Constant	8.090*** (1.222)	12.420** (5.628)	8.175*** (1.226)	1.645*** (0.311)	−5.276 (6.486)	1.587*** (3.310)
Time FE	No	No	Yes	No	No	Yes
Observations	10,353	10,353	10,353	6,481	6,481	6,907
R ²	0.246	0.246	0.247	0.686	0.687	0.687

Note: *p<0.1; **p<0.05; ***p<0.01. Standard errors in parentheses.

Table B.6: Term Structure on the Individual Level - 2009 Q1 - 2024 Q4

	$\mathbf{E}_{i,t}\pi_{t+8}$					
	Households			Business Leaders		
π_t	-0.198*** (0.020)	0.019 (0.025)		0.045*** (0.017)	0.051** (0.020)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.597*** (0.005)	0.595*** (0.005)	0.590*** (0.005)	0.915*** (0.004)	0.915*** (0.004)	0.906*** (0.005)
Female	0.655*** (0.063)	0.621*** (0.063)	0.656*** (0.063)			
Age	-0.018*** (0.002)	-0.019*** (0.002)	-0.019*** (0.002)			
Private_sector	-3.144*** (0.682)	-3.039*** (0.680)	-3.007*** (0.678)			
Public_sector	-2.934*** (0.683)	-2.904*** (0.681)	-2.868*** (0.679)			
Retired	-2.381*** (0.684)	-2.330*** (0.682)	-2.274*** (0.681)			
Student	-1.090 (0.690)	-1.003 (0.688)	-0.901 (0.686)			
Unemployed	-1.827** (0.729)	-1.778** (0.726)	-1.699** (0.725)			
Manufacturing				-0.337*** (0.080)	-0.338*** (0.080)	-0.345*** (0.080)
Retail_trade				-0.299*** (0.084)	-0.297*** (0.084)	-0.303*** (0.083)
Services				-0.443*** (0.073)	-0.435*** (0.073)	-0.419*** (0.073)
Size > 50				-0.198*** (0.050)	-0.193*** (0.050)	-0.111** (0.055)
GDP		0.040 (0.027)			0.013 (0.019)	
I44		-0.058*** (0.004)			-0.003 (0.003)	
Oil		-0.004*** (0.001)			0.0004 (0.001)	
Constant	6.997*** (0.687)	12.426*** (0.772)	8.059*** (0.719)	1.373*** (0.080)	1.665*** (0.277)	1.778*** (0.191)
Time FE	No	No	Yes	No	No	Yes
Observations	40,361	40,361	40,361	24,933	24,933	25,359
R ²	0.276	0.281	0.286	0.655	0.655	0.657

*p<0.1; **p<0.05; ***p<0.01

Table B.7: Term Structure on the Individual Level — 2009 Q1–2021 Q2

	$\mathbf{E}_{i,t}\pi_{t+8}$					
	Economists			Social Partners		
π_t	−0.046*** (0.013)	0.010 (0.013)		−0.130*** (0.026)	−0.082*** (0.027)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.683*** (0.016)	0.710*** (0.016)	0.721*** (0.016)	1.272*** (0.023)	1.297*** (0.023)	1.310*** (0.023)
Financial sector	−0.232*** (0.022)	−0.204*** (0.022)	−0.190*** (0.022)			
Employer organization				0.078* (0.042)	0.081* (0.042)	0.090** (0.041)
GDP		−0.006 (0.008)			−0.023 (0.016)	
I44		−0.017*** (0.001)			−0.015*** (0.003)	
Oil		0.0001 (0.0003)			−0.0004 (0.001)	
Constant	1.058*** (0.042)	2.559*** (0.123)	0.958*** (0.074)	−0.093 (0.074)	1.282*** (0.249)	−0.068 (0.154)
Time FE	No	No	Yes	No	No	Yes
Observations	2,918	2,918	2,918	1,563	1,563	1,563
R ²	0.428	0.463	0.490	0.668	0.675	0.693

*p<0.1; **p<0.05; ***p<0.01

Table B.8: Term Structure on the Individual Level — 2021 Q3–2024 Q4

	$\mathbf{E}_{i,t}\pi_{t+8}$					
	Economists			Social Partners		
π_t	−0.047*** (0.017)	−0.051* (0.027)		−0.228*** (0.039)	−0.212*** (0.062)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.550*** (0.020)	0.556*** (0.022)	0.556*** (0.022)	1.176*** (0.028)	1.200*** (0.028)	1.202*** (0.027)
Financial sector	−0.228*** (0.050)	−0.220*** (0.051)	−0.211*** (0.049)			
Employer organization				0.294*** (0.109)	0.247** (0.107)	0.238** (0.099)
GDP		0.038 (0.038)			0.082 (0.089)	
I44		0.002 (0.016)			0.019 (0.038)	
Oil		−0.0005 (0.003)			0.009 (0.007)	
Constant	1.311*** (0.093)	1.033 (1.982)	1.099*** (0.106)	−0.315 (0.201)	−2.938 (4.711)	−0.606*** (0.228)
Time FE	No	No	Yes	No	No	Yes
Observations	603	603	643	489	489	531
R ²	0.655	0.656	0.664	0.786	0.800	0.807

*p<0.1; **p<0.05; ***p<0.01

Table B.9: Term Structure on the Individual Level - 2009 Q1 - 2024 Q4

	$\mathbf{E}_{i,t}\pi_{t+8}$					
	Economists			Social Partners		
π_t	-0.073*** (0.008)	-0.028*** (0.009)		-0.308*** (0.017)	-0.247*** (0.019)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.614*** (0.012)	0.645*** (0.012)	0.660*** (0.013)	1.167*** (0.016)	1.198*** (0.016)	1.246*** (0.016)
Financial sector	-0.234*** (0.021)	-0.205*** (0.020)	-0.188*** (0.020)			
Employer organization				0.134*** (0.042)	0.133*** (0.042)	0.125*** (0.040)
GDP		0.002 (0.008)			-0.005 (0.018)	
I44		-0.014*** (0.001)			-0.020*** (0.002)	
Oil		0.0002 (0.0003)			0.0003 (0.001)	
Constant	1.250*** (0.027)	2.407*** (0.102)	1.067*** (0.072)	0.418*** (0.053)	2.285*** (0.235)	0.042 (0.167)
Time FE	No	No	Yes	No	No	Yes
Observations	3,521	3,521	3,561	2,052	2,052	2,094
R ²	0.526	0.548	0.571	0.732	0.742	0.766

*p<0.1; **p<0.05; ***p<0.01

Table B.10: Determinants of Wage Growth Expectations - Households - 2009Q1 - 2021Q2

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.228*** (0.007)	0.227*** (0.007)	0.163*** (0.013)	0.228*** (0.007)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.062*** (0.015)	0.063*** (0.015)	0.086*** (0.031)	0.183*** (0.049)	0.069*** (0.012)
$\mathbf{E}_{i,t}\pi_{t+4}$:Female				0.010 (0.032)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Age				-0.003*** (0.001)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Public_sector				0.037 (0.032)	
π_t	-0.053 (0.102)		0.091 (0.241)	-0.047 (0.102)	-0.084 (0.072)
Female	-0.611*** (0.149)	-0.612*** (0.149)	-0.382 (0.271)	-0.630*** (0.169)	-0.655*** (0.117)
Age	-0.051*** (0.006)	-0.051*** (0.006)	-0.057*** (0.011)	-0.042*** (0.006)	-0.026*** (0.003)
Public_sector	-0.564*** (0.150)	-0.570*** (0.151)	-0.396 (0.273)	-0.653*** (0.170)	-0.803*** (0.118)
Oil	0.003 (0.002)		0.008* (0.005)	0.003 (0.002)	0.002 (0.002)
GDP	0.005 (0.051)		-0.266 (0.225)	0.004 (0.051)	0.052 (0.047)
I44	-0.006 (0.020)		0.054 (0.034)	-0.005 (0.020)	-0.012* (0.007)
Constant	5.677** (2.325)	4.394*** (0.462)	-0.612 (3.945)	5.193** (2.331)	6.304*** (0.711)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	11,906	11,906	3,337	11,906	19,379
R ²	0.102	0.103	0.067	0.103	0.013

*p<0.1; **p<0.05; ***p<0.01

Table B.11: Determinants of Wage Growth Expectations - Households - 2021Q3 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.067*** (0.008)	0.067*** (0.008)	0.040** (0.020)	0.065*** (0.008)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.052*** (0.013)	0.050*** (0.013)	0.086*** (0.033)	0.279*** (0.041)	0.049*** (0.013)
$\mathbf{E}_{i,t}\pi_{t+4}$:Female				-0.004 (0.028)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Age				-0.005*** (0.001)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Public_sector				-0.010 (0.029)	
CPI	-0.286** (0.112)			-0.284** (0.112)	-0.297*** (0.113)
Female	-0.645*** (0.209)	-0.639*** (0.209)	-0.345 (0.500)	-0.619** (0.248)	-0.761*** (0.210)
Age	-0.057*** (0.007)	-0.057*** (0.007)	-0.083*** (0.018)	-0.031*** (0.009)	-0.070*** (0.007)
Public_sector	-0.983*** (0.210)	-0.964*** (0.210)	-1.437*** (0.504)	-0.938*** (0.251)	-1.108*** (0.211)
Oil	-0.009 (0.012)		0.027 (0.033)	-0.009 (0.012)	-0.008 (0.012)
GDP	0.005 (0.158)		-0.412 (0.988)	0.014 (0.158)	-0.008 (0.159)
I44	-0.030 (0.068)			-0.034 (0.068)	-0.037 (0.069)
Constant	11.651 (8.417)	6.343*** (0.503)	8.381*** (1.269)	10.978 (8.400)	13.572 (8.488)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	7,058	7,058	1,534	7,058	7,427
R ²	0.033	0.035	0.034	0.037	0.025

*p<0.1; **p<0.05; ***p<0.01

Table B.12: Determinants of Wage Growth Expectations - Business Leaders - 2009Q1 - 2021Q2

	$E_{i,t}w_{t+4}$				
$E_{i,t}\pi_{t+4}$	0.305*** (0.005)	0.229*** (0.005)	0.073*** (0.004)	0.470*** (0.012)	0.467*** (0.005)
$E_{i,t}\pi_{t+4}$:Size > 50				-0.051*** (0.010)	
$E_{i,t}\pi_{t+4}$:Manufacturing				-0.105*** (0.015)	
$E_{i,t}\pi_{t+4}$:Retail_trade				-0.077*** (0.016)	
$E_{i,t}\pi_{t+4}$:Services				-0.221*** (0.013)	
π_t	0.073*** (0.018)		0.012 (0.014)	0.085*** (0.018)	-0.026 (0.021)
$E_{i,t}w_t$	0.354*** (0.005)	0.550*** (0.006)	2.964*** (0.023)	0.357*** (0.005)	
Size > 50	-0.011 (0.029)	-0.021 (0.032)	0.073*** (0.018)	0.138*** (0.039)	0.026 (0.033)
Manufacturing	-0.098** (0.047)	-0.076* (0.044)	-0.002 (0.029)	0.212*** (0.062)	-0.065 (0.053)
Retail_trade	-0.081 (0.050)	-0.071 (0.047)	-0.001 (0.031)	0.151** (0.067)	-0.043 (0.057)
Services	0.039 (0.044)	0.003 (0.041)	0.075*** (0.027)	0.662*** (0.056)	0.107** (0.049)
Oil	0.002*** (0.0004)		0.008*** (0.0003)	0.002*** (0.0004)	0.001 (0.0005)
GDP	-0.011 (0.011)		0.185*** (0.007)	-0.012 (0.010)	0.011 (0.012)
I44	-0.014*** (0.002)		0.047*** (0.001)	-0.015*** (0.002)	-0.032*** (0.002)
Constant	2.086*** (0.177)	1.171*** (0.094)	-6.978*** (0.142)	1.704*** (0.178)	4.978*** (0.194)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	18,629	18,629	5,174	18,629	18,629
R ²	0.448	0.523	0.788	0.460	0.302

*p<0.1; **p<0.05; ***p<0.01

Table B.13: Determinants of Wage Growth Expectations - Business Leaders - 2021Q3 - 2024Q4

	$E_{i,t}w_{t+4}$				
$E_{i,t}\pi_{t+4}$	0.329*** (0.007)	0.206*** (0.006)	0.050*** (0.005)	0.230*** (0.016)	0.625*** (0.008)
$E_{i,t}\pi_{t+4}$:Size > 50				0.078*** (0.013)	
$E_{i,t}\pi_{t+4}$:Manufacturing				0.001 (0.016)	
$E_{i,t}\pi_{t+4}$:Retail_trade				0.063*** (0.019)	
$E_{i,t}\pi_{t+4}$:Services				0.083*** (0.016)	
π_t	-0.074 (0.061)			-0.067 (0.060)	-0.161* (0.083)
$E_{i,t}w_t$	0.761*** (0.010)	1.084*** (0.010)	3.828*** (0.023)	0.769*** (0.010)	
Size > 50	0.200* (0.109)	0.100 (0.085)	-0.021 (0.045)	-0.333** (0.139)	0.328** (0.148)
Manufacturing	-0.033 (0.168)	-0.099 (0.132)	0.062 (0.069)	-0.036 (0.208)	0.500** (0.230)
Retail_trade	-0.039 (0.169)	0.023 (0.133)	0.084 (0.069)	-0.482** (0.217)	-0.040 (0.231)
Services	0.241 (0.149)	0.209* (0.117)	0.054 (0.061)	-0.346* (0.187)	0.417** (0.203)
Oil	-0.030*** (0.007)		-0.061*** (0.003)	-0.029*** (0.007)	0.004 (0.009)
GDP	0.282*** (0.085)		2.235*** (0.099)	0.292*** (0.085)	-0.001 (0.117)
I44	-0.205*** (0.037)			-0.205*** (0.037)	0.098* (0.051)
Constant	21.860*** (4.581)	-3.310*** (0.175)	-6.684*** (0.119)	22.457*** (4.554)	-9.946 (6.230)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	6,203	6,203	1,372	6,203	6,203
R ²	0.738	0.839	0.983	0.741	0.510

*p<0.1; **p<0.05; ***p<0.01

Table B.14: Determinants of Wage Growth Expectations - Economists - 2009Q1 - 2021Q2

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.081*** (0.007)	1.070*** (0.019)	1.899*** (0.059)	0.081*** (0.007)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.221*** (0.020)	0.083*** (0.014)	0.161*** (0.025)	0.219*** (0.024)	0.231*** (0.020)
$\mathbf{E}_{i,t}\pi_{t+4}:\text{ec_fi}$				0.005 (0.042)	
π_t	-0.055*** (0.017)		-0.182*** (0.024)	-0.055*** (0.017)	-0.091*** (0.017)
ec_fi	0.121*** (0.028)	0.017 (0.019)	0.113*** (0.034)	0.109 (0.090)	0.123*** (0.028)
Oil	0.001** (0.0004)		0.005*** (0.001)	0.001** (0.0004)	0.001 (0.0004)
GDP	-0.009 (0.010)		0.069*** (0.013)	-0.009 (0.010)	-0.009 (0.010)
I44	-0.052*** (0.002)		0.017*** (0.003)	-0.052*** (0.002)	-0.057*** (0.002)
Constant	7.523*** (0.173)	0.716*** (0.076)	-1.858*** (0.360)	7.527*** (0.176)	8.425*** (0.159)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	2,911	2,911	743	2,911	2,911
R ²	0.389	0.725	0.727	0.389	0.360

*p<0.1; **p<0.05; ***p<0.01

Table B.15: Determinants of Wage Growth Expectations - Economists - 2021Q3 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	-0.007 (0.010)	0.821*** (0.045)	3.535*** (0.078)	-0.008 (0.010)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.233*** (0.026)	0.140*** (0.020)	0.023 (0.015)	0.245*** (0.030)	0.232*** (0.025)
$\mathbf{E}_{i,t}\pi_{t+4}:\text{ec_fi}$				-0.033 (0.045)	
π_t	-0.041 (0.031)			-0.041 (0.031)	-0.040 (0.031)
ec.fi	0.273*** (0.059)	0.103** (0.045)	-0.022 (0.030)	0.386** (0.168)	0.271*** (0.059)
Oil	-0.008** (0.003)		-0.060*** (0.003)	-0.008** (0.003)	-0.008** (0.003)
GDP	-0.018 (0.044)		2.143*** (0.094)	-0.019 (0.044)	-0.015 (0.044)
I44	0.010 (0.019)			0.011 (0.019)	0.007 (0.019)
Constant	2.507 (2.335)	-1.303*** (0.225)	-5.759*** (0.242)	2.342 (2.348)	2.760 (2.306)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	610	610	131	610	610
R ²	0.378	0.656	0.963	0.379	0.378
Adjusted R ²	0.371	0.647	0.962	0.370	0.372
Residual Std. Error	0.641	0.480	0.145	0.641	0.640
F Statistic	52.313***	70.751***	650.972***	45.802***	61.003***

*p<0.1; **p<0.05; ***p<0.01

Table B.16: Determinants of Wage Growth Expectations - Social Partners - 2009Q1 - 2021Q2

	$E_{i,t}w_{t+4}$				
$E_{i,t}w_t$	0.046*** (0.009)	0.989*** (0.049)	1.465*** (0.069)	0.046*** (0.009)	
$E_{i,t}\pi_{t+4}$	0.116*** (0.019)	0.066*** (0.016)	0.116*** (0.032)	0.147*** (0.043)	0.120*** (0.019)
$E_{i,t}\pi_{t+4}:\text{sp_er}$				-0.038 (0.047)	
π_t	-0.094*** (0.023)		-0.136*** (0.026)	-0.097*** (0.023)	-0.118*** (0.022)
sp_er	-0.225*** (0.034)	-0.111*** (0.029)	-0.164*** (0.034)	-0.138 (0.112)	-0.232*** (0.034)
Oil	0.001 (0.001)		0.004*** (0.001)	0.001 (0.001)	0.001 (0.001)
GDP	-0.036*** (0.013)		0.006 (0.015)	-0.036*** (0.013)	-0.034** (0.013)
I44	-0.048*** (0.002)		0.004 (0.003)	-0.048*** (0.002)	-0.051*** (0.002)
Constant	7.880*** (0.226)	1.337*** (0.162)	0.494 (0.393)	7.831*** (0.234)	8.374*** (0.205)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	1,551	1,551	430	1,551	1,551
R ²	0.359	0.556	0.742	0.359	0.348

*p<0.1; **p<0.05; ***p<0.01

Table B.17: Determinants of Wage Growth Expectations - Social Partners - 2021Q3 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.057*** (0.016)	0.939*** (0.039)	3.838*** (0.088)	0.056*** (0.016)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.189*** (0.022)	0.092*** (0.015)	0.050*** (0.010)	0.143*** (0.043)	0.195*** (0.022)
$\mathbf{E}_{i,t}\pi_{t+4}:\text{sp_er}$				0.060 (0.048)	
π_t	0.017 (0.049)			0.022 (0.049)	0.007 (0.049)
sp_er	-0.233*** (0.083)	-0.168*** (0.055)	0.078* (0.039)	-0.486** (0.218)	-0.230*** (0.084)
Oil	-0.014*** (0.005)		-0.071*** (0.003)	-0.015*** (0.005)	-0.013** (0.005)
GDP	-0.034 (0.070)		2.542*** (0.111)	-0.040 (0.070)	-0.050 (0.070)
I44	-0.054* (0.030)			-0.055* (0.030)	-0.036 (0.030)
Constant	9.731*** (3.698)	-1.685*** (0.218)	-6.996*** (0.289)	10.123*** (3.708)	8.072** (3.713)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	485	485	103	485	485
R ²	0.310	0.705	0.964	0.313	0.292

*p<0.1; **p<0.05; ***p<0.01

Table B.18: Determinants of Wage Growth Expectations - Households - 2009Q1 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.161*** (0.005)	0.160*** (0.005)	0.120*** (0.011)	0.160*** (0.005)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.055*** (0.010)	0.055*** (0.010)	0.093*** (0.022)	0.216*** (0.031)	0.066*** (0.009)
$\mathbf{E}_{i,t}\pi_{t+4}$:Female				0.00002 (0.021)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Age				-0.004*** (0.001)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Public_sector				0.005 (0.021)	
π_t	-0.028 (0.042)		0.037 (0.092)	-0.025 (0.042)	-0.014 (0.040)
Female	-0.628*** (0.122)	-0.624*** (0.122)	-0.389 (0.243)	-0.623*** (0.140)	-0.704*** (0.103)
Age	-0.053*** (0.005)	-0.053*** (0.005)	-0.065*** (0.009)	-0.039*** (0.005)	-0.031*** (0.002)
Public_sector	-0.717*** (0.123)	-0.707*** (0.123)	-0.689*** (0.244)	-0.734*** (0.141)	-0.896*** (0.103)
Oil	0.003* (0.002)		0.010** (0.005)	0.003** (0.002)	0.002 (0.002)
GDP	0.040 (0.047)		0.011 (0.169)	0.040 (0.047)	0.083* (0.043)
I44	0.047*** (0.010)		0.054** (0.025)	0.047*** (0.010)	0.013** (0.006)
Constant	0.454 (1.122)	4.897*** (0.435)	-0.040 (2.689)	-0.128 (1.126)	3.990*** (0.589)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	18,964	18,964	4,871	18,964	26,806
R ²	0.069	0.072	0.053	0.070	0.016

*p<0.1; **p<0.05; ***p<0.01

Table B.19: Determinants of Wage Growth Expectations - Business Leaders - 2009Q1 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.537*** (0.005)	0.850*** (0.005)	3.161*** (0.017)	0.535*** (0.005)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.361*** (0.004)	0.256*** (0.004)	0.109*** (0.004)	0.329*** (0.009)	0.590*** (0.004)
$\mathbf{E}_{i,t}\pi_{t+4}$:Size > 50				0.061*** (0.007)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Manufacturing				0.002 (0.009)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Retail_trade				0.005 (0.011)	
$\mathbf{E}_{i,t}\pi_{t+4}$:Services				-0.031*** (0.009)	
π_t	-0.026* (0.015)		-0.322*** (0.011)	-0.021 (0.015)	-0.063*** (0.018)
Size > 50	0.050 (0.036)	0.032 (0.035)	0.057** (0.023)	-0.172*** (0.044)	0.104** (0.044)
Manufacturing	-0.008 (0.058)	-0.021 (0.050)	0.082** (0.036)	-0.020 (0.068)	0.097 (0.071)
Retail_trade	-0.078 (0.061)	-0.056 (0.052)	0.033 (0.038)	-0.104 (0.074)	-0.040 (0.074)
Services	0.125** (0.053)	0.043 (0.046)	0.143*** (0.033)	0.234*** (0.063)	0.172*** (0.065)
Oil	0.0004 (0.001)		0.006*** (0.0004)	0.0003 (0.001)	-0.002** (0.001)
GDP	-0.028* (0.014)		0.170*** (0.009)	-0.025* (0.014)	0.030* (0.017)
I44	-0.019*** (0.002)		0.037*** (0.002)	-0.019*** (0.002)	-0.010*** (0.003)
Constant	1.895*** (0.201)	0.382*** (0.120)	-5.778*** (0.147)	1.981*** (0.202)	2.435*** (0.245)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	24,832	24,832	6,546	24,832	24,832
R ²	0.648	0.740	0.918	0.649	0.475

*p<0.1; **p<0.05; ***p<0.01

Table B.20: Determinants of Wage Growth Expectations - Economists - 2009Q1 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.143*** (0.006)	1.030*** (0.017)	1.860*** (0.040)	0.143*** (0.006)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.335*** (0.017)	0.101*** (0.011)	0.159*** (0.020)	0.320*** (0.019)	0.408*** (0.018)
$\mathbf{E}_{i,t}\pi_{t+4} \cdot \text{ec_fi}$				0.060** (0.030)	
π_t	0.100*** (0.013)		-0.100*** (0.017)	0.100*** (0.013)	0.086*** (0.015)
ec.fi	0.183*** (0.029)	0.028 (0.018)	0.112*** (0.032)	0.046 (0.074)	0.210*** (0.031)
Oil	0.001** (0.0004)		0.004*** (0.0005)	0.001** (0.0004)	0.001** (0.0005)
GDP	0.001 (0.011)		0.070*** (0.012)	0.001 (0.011)	0.009 (0.012)
I44	-0.030*** (0.002)		0.011*** (0.002)	-0.030*** (0.002)	-0.031*** (0.002)
Constant	4.618*** (0.146)	0.781*** (0.074)	-1.363*** (0.231)	4.663*** (0.148)	5.137*** (0.156)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	3,521	3,521	874	3,521	3,521
R ²	0.320	0.759	0.777	0.320	0.206

*p<0.1; **p<0.05; ***p<0.01

Table B.21: Determinants of Wage Growth Expectations - Social Partners - 2009Q1 - 2024Q4

	$\mathbf{E}_{i,t}w_{t+4}$				
$\mathbf{E}_{i,t}w_t$	0.147*** (0.008)	0.957*** (0.029)	1.699*** (0.046)	0.146*** (0.008)	
$\mathbf{E}_{i,t}\pi_{t+4}$	0.216*** (0.015)	0.081*** (0.011)	0.094*** (0.017)	0.271*** (0.026)	0.260*** (0.016)
$\mathbf{E}_{i,t}\pi_{t+4}:\text{sp_er}$				-0.072*** (0.028)	
π_t	0.178*** (0.017)		-0.015 (0.020)	0.168*** (0.017)	0.173*** (0.018)
sp_er	-0.232*** (0.037)	-0.127*** (0.026)	-0.154*** (0.035)	-0.037 (0.084)	-0.255*** (0.040)
Oil	0.0004 (0.001)		0.004*** (0.001)	0.0004 (0.001)	0.0002 (0.001)
GDP	-0.042*** (0.016)		0.030* (0.015)	-0.040** (0.016)	-0.029* (0.017)
I44	-0.026*** (0.002)		0.006** (0.003)	-0.026*** (0.002)	-0.023*** (0.002)
Constant	4.588*** (0.209)	1.396*** (0.127)	-0.372 (0.272)	4.515*** (0.211)	4.812*** (0.226)
Time FE	No	Yes	No	No	No
Period	All	All	Q1	All	All
Observations	2,036	2,036	533	2,036	2,036
R ²	0.343	0.701	0.811	0.346	0.229

*p<0.1; **p<0.05; ***p<0.01

Table B.22: Inflation-wage Pass-through

	w_{t+1}		
w_t	0.70*** (0.16)	0.70*** (0.16)	0.69***
π_{t+1}		0.17 (0.13)	
π_t			0.12 (0.13)
Observations	22	22	22
R^2	0.47	0.49	0.47

*p<0.1; **p<0.05; ***p<0.01

Note: Regression of actual wage growth, w_{t+1} , on lagged wage growth, w_t , contemporaneous inflation, π_{t+1} , or past inflation, π_t , annual frequency, 2000-2024.

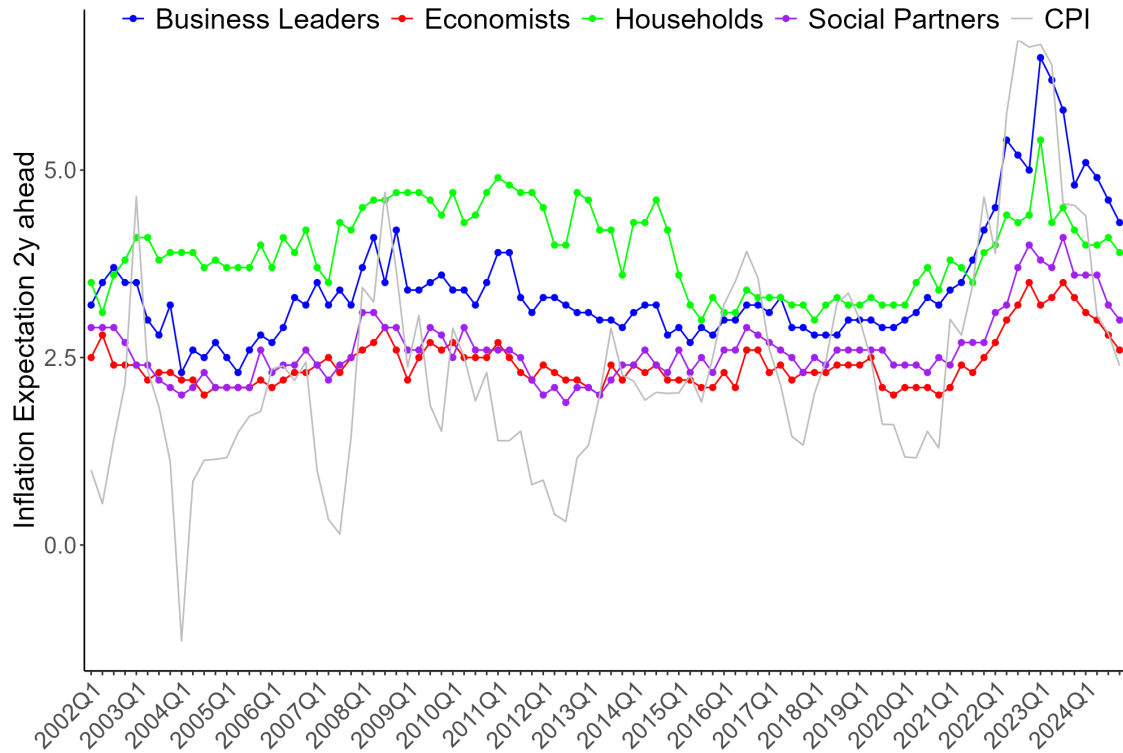
Table B.23: Effect of Energy Prices on Inflation Expectations

	Firms		Households	
	$E_{i,r,s,t}\pi_{t+4}$	$E_{i,r,s,t}\pi_{t+8}$	$E_{i,r,s,t}\pi_{t+4}$	$E_{i,r,s,t}\pi_{t+8}$
$D_{r,t}$	1.011*** (0.179)	0.947*** (0.211)	0.086 (0.152)	0.064 (0.202)
$D_{r,t} \times \Delta P$	0.0002 (0.002)	0.005** (0.002)	-0.002 (0.002)	-0.001 (0.002)
Time FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	42,973	41,603	83,031	73,323
R^2	0.175	0.131	0.045	0.037

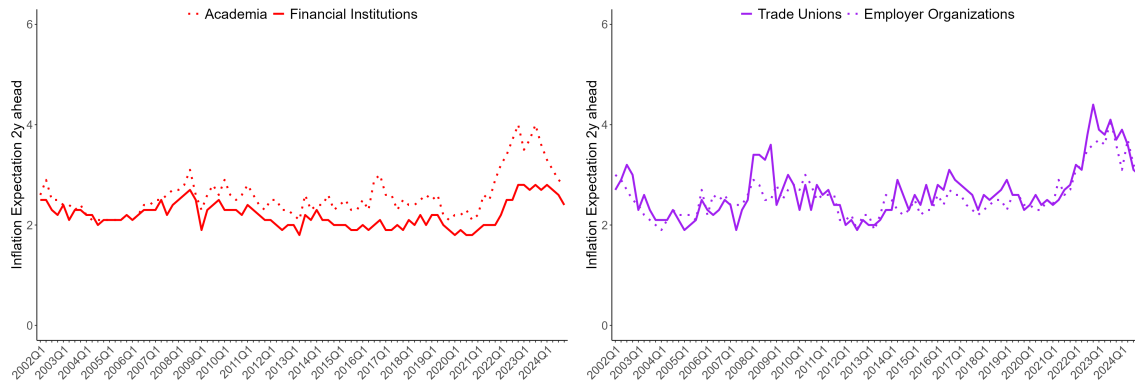
*p<0.1; **p<0.05; ***p<0.01

Note: Regression from model (3) for business leaders and households. All models include time fixed effects, a dummy for living in the south and individual-level controls. For households, controls include age, sex and occupational status, while sector and size for business leaders.

C Figures



a All Agents



b Economists and Social Partners

Figure C.1: Timeseries of average inflation expectations (2 years ahead) since 2002
Note: The bottom left hand side graph plots academics and financial institutions, the bottom right hand side graph plots employer organizations and trade unions. The black line in (a) indicates CPI inflation in Norway.

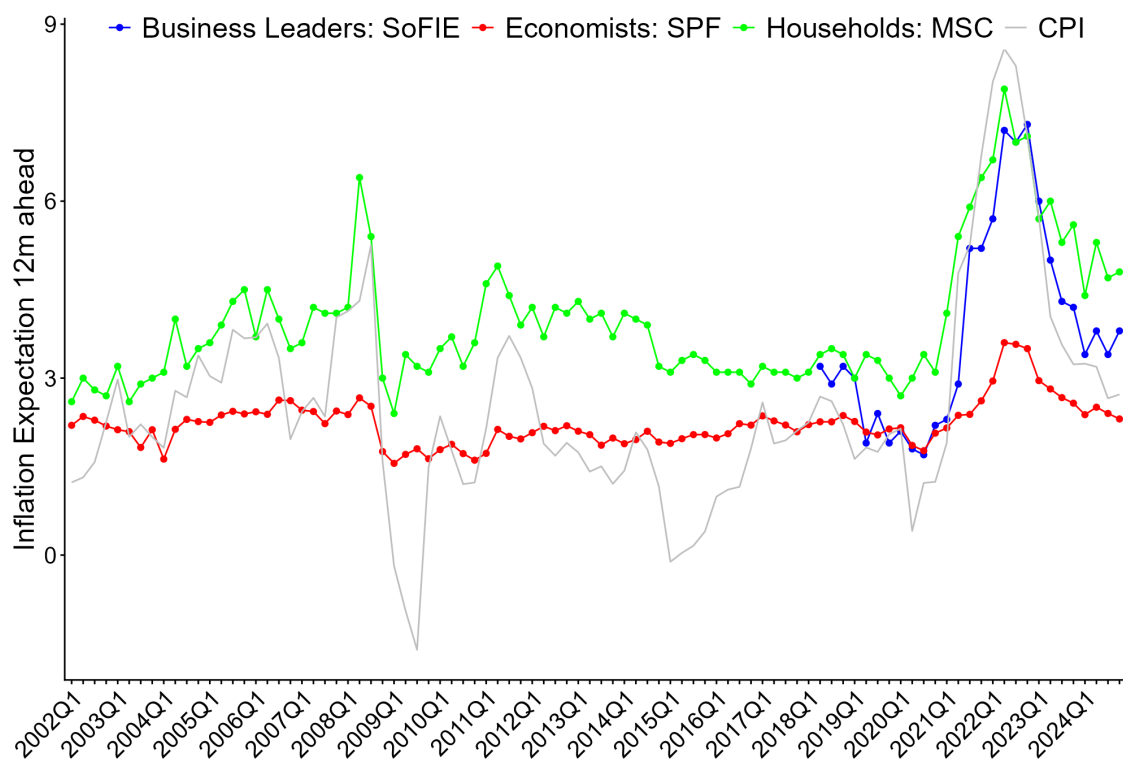


Figure C.2: Timeseries of average inflation expectations (12 months ahead) and year-over-year CPI inflation since 2002 in the US.

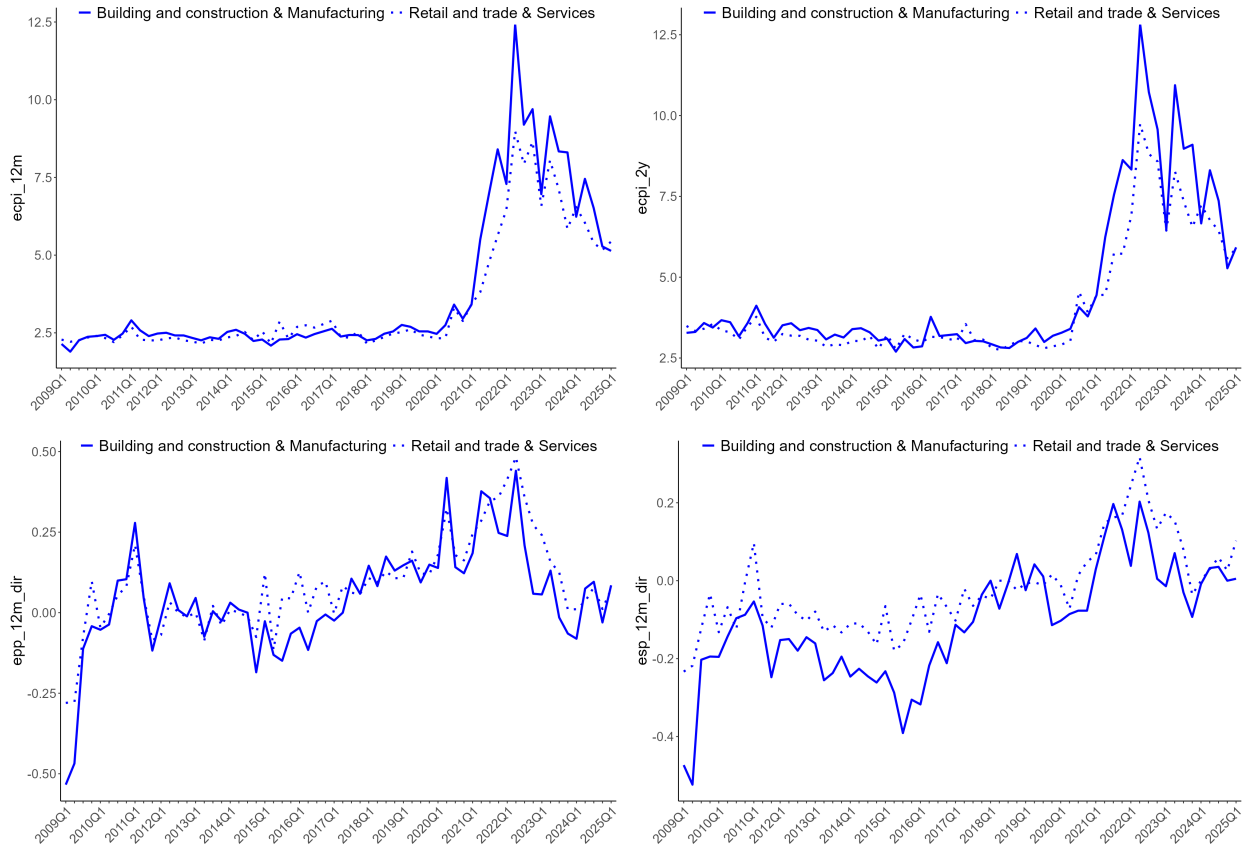


Figure C.3: Timeseries of business leaders' expectations by sector since 2012

Note: The top left hand side graphs plot short-term expectations (over the next 12 months) and the top right hand side graphs plot medium-term expectations (over the next 2 years). The bottom left hand side graphs plot the balance statistic for purchasing prices, the bottom right hand side or sales prices.

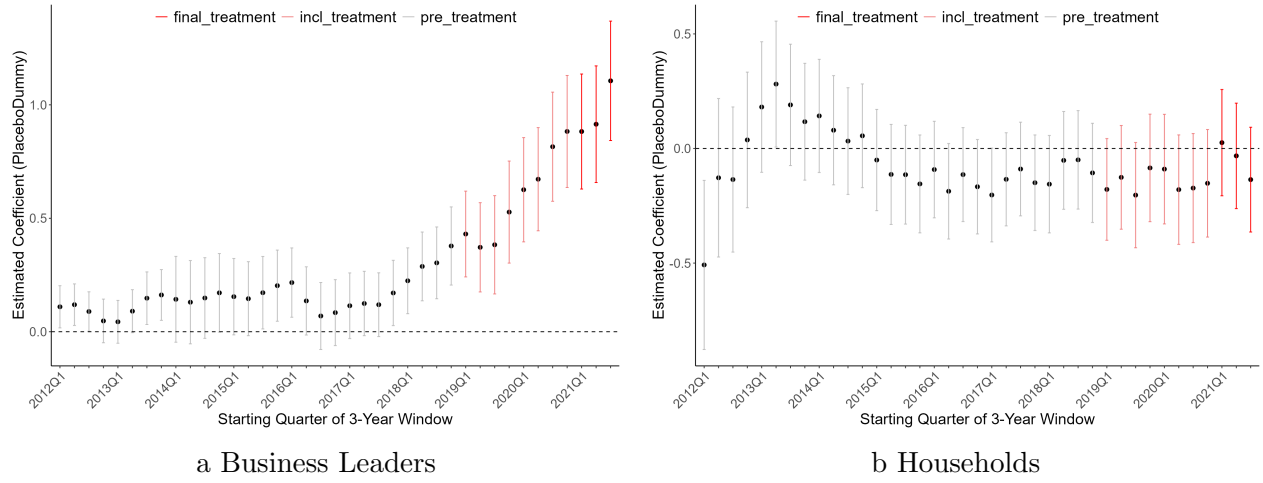


Figure C.4: Placebo Test Using Rolling 3 Year Treatment Windows

Note: The placebo test re-estimates the main regression (3) using moving 3-year windows (keeping the first 3 years as baseline) as artificial treatment periods. For each window, a placebo dummy is defined for the treated region, and the resulting coefficients are plotted together with the 95% confidence interval to check for spurious effects outside the actual intervention period. The gray windows are estimated over the pre-treatment period, the pink windows include some months of the treatment period and the red windows include only our final treatment window in the dummy.